NEET AIR (2027)

[Special Test Series]

TEST-01

DURATION: 180 MIN. | **DATE : 13/08/2026** | **M. MARKS: 720**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Botany:	Full Syllabus
Zoology:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **720 minutes** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. **45** questions in each subject
3. The test booklet consists of **180** questions. The maximum marks are **720**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics, 1-45)**, Section-II **(Chemistry, 46-90)**, Section-III **(Botany, 91-135)**, Section-IV **(Zoology, 136-180)** and having **45** questions in each part in which all **45** questions are of Objective Type and all **45** questions are **compulsory.**
5. There is only **one correct** response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
8. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. Use of white fluid for correction is **not permissible** on the **Answer Sheet**.
9. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

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OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the **OMR Sheet** is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

Q1 A man standing on the roof of a house of height *h* throws one particle vertically downwards with speed '*u*' and another particle horizontally with the same speed *u*. The ratio of their speeds just before when they reach the earth's surface will be: (Neglect air resistance)

(1) $\sqrt{2gh+u^2} : u$ (2) 1 : 2

(3) 1 : 1 (4) $\sqrt{2gh+u^2} : \sqrt{2gh}$

Q2 A potential difference of 20V is applied across a conductor of length 10 cm. If drift velocity of electrons is 2 × 10^{-4} m/s, then electron mobility will be:

(1) 1 × 10^{-6} m^2 V^{-1} s^{-1}

(2) 2 × 10^{-6} m^2 V^{-1} s^{-1}

(3) 3 × 10^{-6} m^2 V^{-1} s^{-1}

(4) 4 × 10^{-6} m^2 V^{-1} s^{-1}

Q3 The velocity of a particle moving in a straight line varies with time as $v(t) = (t^2 - 5t + 6)\text{m/s}$. (where *t* is in second). Which of the following statement is **correct** for $\frac{5}{2}\text{s} < t < 3s$?

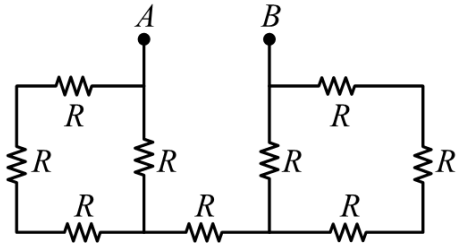
(1) The particle is speeding up

(2) The particle is slowing down

(3) The speed is constant

(4) Data insufficient

Q4 The equivalent resistance of the circuit across *A* and *B* as shown in the figure is;



(1) $\frac{5R}{2}$ (2) $\frac{3R}{4}$

(3) $\frac{5R}{3}$ (4) $\frac{2R}{5}$

Q5 A particle moves along the x–axis from $x = 1$ to $x = e$ by a force $F(x) = k/x$ (Where *k* is positive constant). The work done is W_1. The same particle moves from $x = e$ to $x = e^2$ under the same force. The work done is W_2. What is the ratio $W_1 : W_2$? (e stands for the exponential function)

(1) 1 : e^2 (2) 1 : e

(3) 1 : 2 (4) 1 : 1

Q6 The magnetic energy stored in a current carrying coil having inductance *L* and resistance *R* is *U*. Now this coil is divided into four equal parts and connected in parallel to same voltage source. The total energy stored in the combination becomes

(1) $\frac{U}{2}$ (2) $\frac{U}{4}$

(3) $\frac{U}{16}$ (4) 16 *U*

Q7 Two boys are holding a horizontal uniform rod of length *L* and weight *W* through its two ends. If now one of the boys suddenly leaves the rod, what is the instantaneous reaction force experienced by the other boy?

(1) $\frac{W}{2}$ (2) $\frac{W}{4}$

(3) $\frac{2W}{3}$ (4) $\frac{W}{3}$

Q8 Match **List-I** (devices) with **List-II** (their applications)

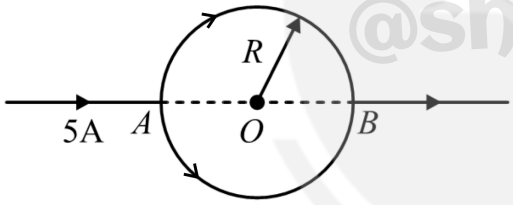
List-I		List-II	
(A)	A.C. generator	(I)	Detect the presence of small current in circuit.
(B)	Light emitting diode	(II)	Change input A.C voltage to higher or smaller value
(C)	Transformer	(III)	Convert mechanical energy into electrical energy
(D)	Galvanometer	(IV)	Convert electrical energy into light energy

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(1) A-IV, B-III, C-II, D-I
(2) A-I, B-II, C-III, D-IV
(3) A-IV, B-II, C-III, D-I
(4) A-III, B-IV, C-II, D-I

Q9 A student wants to derive a formula for the orbital time period T of a satellite just above the Earth's surface in terms of the Earth's density d and the gravitational constant G. Which of the following information is necessary?
(1) Radius of the Earth only
(2) Mass of the Earth only
(3) Density of the Earth and the gravitational constant
(4) Density of the Earth, gravitational constant and radius of the Earth

Q10 The current flowing in a straight long conductor is 5A. At point A the current divides into two semicircular loops of radius 1m. The electric resistance of one of the semicircle is 3Ω and that of the other is 2Ω. What is the net magnetic field at the common centre O?

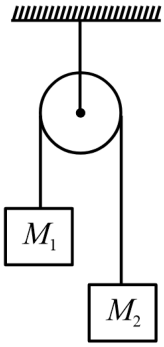


(1) Zero
(2) $\pi \times 10^{-7}$ T
(3) $2\pi \times 10^{-7}$ T
(4) $\frac{\pi}{2} \times 10^{-7}$ T

Q11 The wavelength of maximum intensity of radiation emitted by a star is 500 nm. What is the temperature of the star? (Assume the star acts as a black body)
(1) 5800 K
(2) 2900 K
(3) 11600 K
(4) 4500 K

Q12 Two point charges A and B, having charges $+Q$ and $-Q$ respectively, are placed at a certain distance apart and force acting between them is F. If 20% charge of A is transferred to B, then force between the charges become:
(1) $\frac{9F}{16}$ (2) $\frac{4F}{5}$
(3) $\frac{16F}{25}$ (4) $\frac{3F}{4}$

Q13 Two masses M_1 and M_2 are tied together at the two ends of a light inextensible string that passes over a frictionless pulley. When the mass M_2 is twice that of M_1, the acceleration of the system is a_1. When the mass M_2 is thrice that of M_1, the acceleration of system is a_2. The ratio $\frac{a_1}{a_2}$ will be:



(1) $\frac{1}{3}$ (2) $\frac{2}{3}$
(3) $\frac{3}{2}$ (4) $\frac{2}{1}$

Q14 Three capacitors of capacitances 1 μF, 2 μF and 3 μF are connected in series and a potential difference of 22 V is applied across the combination of the capacitors. Then, the charge that will appear on the positive plate of 3 μF capacitor in steady state will be:
(1) 132 μC (2) 22 μC
(3) 11 μC (4) 12 μC

Q15 In the oscillation of simple pendulum near the surface of Earth, the restoring torque required for the oscillation about the point of suspension is provided by:
(1) tension in the string
(2) centripetal force
(3) one component of weight of the bob
(4) no restoring torque is present

Q16 The focal length of a concave mirror is 30 cm. What is the position of the real object placed in front of mirror such that virtual image is three times the size of the object?
(1) 20 cm (2) 30 cm

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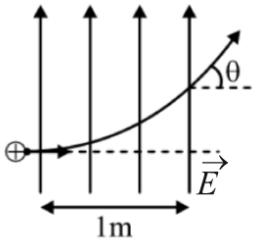
(3) 40 cm (4) 10 cm

Q17 A sonometer wire of length 114 cm is fixed at both ends. Where should the two bridges are placed so that to divide the wire into three segments whose fundamental frequencies are in the ratio 1 : 3 : 4?
(1) At 72 cm and 96 cm from one end.
(2) At 72 cm and 18 cm from one end.
(3) At 36 cm and 84 cm from one end.
(4) At 24 cm and 72 cm from one end.

Q18 Light from a point source in air falls on a convex spherical glass surface having refractive index 1.5 and radius of curvature = 40 cm. If image is formed at infinity, then distance of the light source from the glass surface is:
(1) 20 cm (2) 40 cm
(3) 80 cm (4) Infinity

Q19 A Carnot engine operating between a source temperature T_1 and a sink temperature T_2 has an efficiency of 20%. When the temperature of the sink is reduced by 40 K, the efficiency of the engine increases to 40%. The temperature of the source (T_1) is:
(1) 100 K (2) 200 K
(3) 300 K (4) 400 K

Q20 What is the angle of deviation (θ) for a positive charged particle having charge 2 μC that enters a uniform electric field E = 10 V/m, perpendicularly as shown with speed 5 m/s? (Take the mass of charged particle as 1 mg and ignore gravitational force)

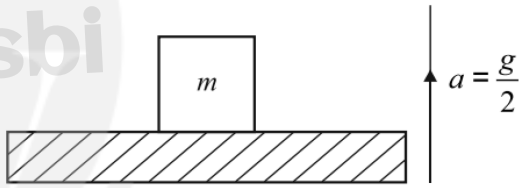


(1) 60° (2) $\tan^{-1}\left(\frac{4}{5}\right)$
(3) 45° (4) $\tan^{-1}\left(\frac{4}{3}\right)$

Q21 A man walking horizontally at 4 m/s observes that rain appears to be falling at an angle of 30° to the vertical. When he increases his speed to 8 m/s, he observes that the rain now appears to be falling at an angle of 60° to the vertical. What is the actual speed of the rain with respect to the ground?
(1) 4 m/s (2) $4\sqrt{3}$ m/s
(3) $\frac{8}{\sqrt{3}}$ m/s (4) 6 m/s

Q22 Two point charges each having 2μC charge are placed on the x-axis at x = 0 and x = 3 cm. Keeping the charge at x = 0 as fixed and the other charge is moved slowly to a point (6 cm, 8cm) in the xy plane. What is the magnitude of change in electrostatic potential energy of the system of two charges?
(1) 0.42 J (2) 0.36 J
(3) 0.84 J (4) 1.20 J

Q23 A block of mass m is kept on a platform which starts from rest with constant acceleration $g/2$ upward, as shown in figure. Work done by normal reaction on block in time t is



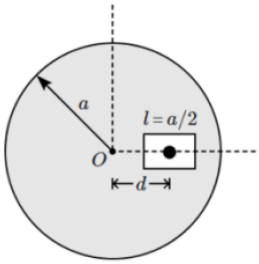
(1) $-\frac{mg^2t^2}{8}$
(2) $\frac{mg^2t^2}{8}$
(3) 0
(4) $\frac{3mg^2t^2}{8}$

Q24 Three prisms A, B and C each having the prism angle 6° but their refractive indices are respectively 1.4, 1.6 and 1.8. If $\delta_A, \delta_B, \delta_C$ be their respective angles of deviation then:
(1) $\delta_A > \delta_B > \delta_C$
(2) $\delta_A < \delta_B = \delta_C$
(3) $\delta_A < \delta_B < \delta_C$
(4) $\delta_A = \delta_B > \delta_C$

Q25 A square shaped hole of side $l = \frac{a}{2}$ is carved out at a distance $d = \frac{a}{2}$ from the centre 'O' of a uniform circular disc of radius a. If the distance of the centre of mass of the remaining portion from

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O is $-\frac{a}{x}$, then the value of x (to the nearest integer) is:



(1) 12 (2) 23
(3) 45 (4) 76

Q26 The ratio of the de-Broglie wavelength associated with proton and electron having same kinetic energy is: ($m_p = m_e \times 1849$)
(1) 1 : 43 (2) 1 : 62
(3) 1 : 30 (4) 2 : 43

Q27 A metal rod of length 1 m and diameter 2 cm is stretched slightly along its length within the elastic limit. The longitudinal strain produced is 0.004. The Poisson's ratio of the material is 0.30. What is the magnitude of lateral strain in the rod?
(1) 4.0×10^{-4}
(2) 1.2×10^{-3}
(3) 3.0×10^{-3}
(4) 7.5×10^{-4}

Q28 Given below are two statements:
Statement I: Out of microwaves, infrared rays and ultraviolet rays, ultraviolet rays are the most likely to be effective for the emission of electrons from a metallic surface.
Statement II: If the incident light frequency is above the threshold frequency, the maximum kinetic energy of photoelectrons is inversely proportional to the frequency of the incident light.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

Q29 In a hydraulic lift, the input piston has a surface area of 5 cm^2 and the output piston has a surface area of 100 cm^2. If a force of 50 N is applied to the input piston, how much mass can the output piston lift? (Take g = 10 m/s^2) (Assume piston is massless and friction between piston and wall is neglected)
(1) 50 kg (2) 100 kg
(3) 200 kg (4) 10 kg

Q30 The magnetic flux across a loop of resistance 10Ω is given by $\phi = (5t^2 - 4t + 1)$ Wb. The charge flown in the loop from $t = 0$ to $t = 2$ s is:
(1) 1 C (2) 1.2 C
(3) 0.6 C (4) 2.4 C

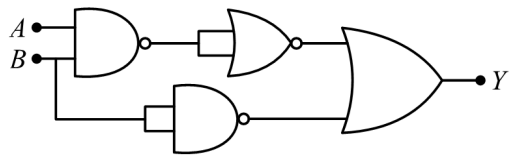
Q31 A certain mass of an ideal gas is contained in a rigid vessel of volume V, exerting a pressure P on the walls. If the gas is transferred to another vessel of volume $4V$ and it is ensured that the root mean square speed (v_{rms}) of the gas molecules remains the same as before, the new pressure exerted by the gas will be:
(1) $4P$ (2) $\frac{P}{4}$
(3) $16P$ (4) $\frac{P}{2}$

Q32 When a substance is placed near a bar magnet it is weakly attracted towards it, then the relative magnetic permeability of the substance can be:
(1) 1.1 (2) 0.80
(3) 500 (4) 0.5

Q33 A block attached to a spring executes SHM with an amplitude of 25 cm. At a certain position where the displacement is $y = A\cos(37^\circ)$ units, the kinetic energy of the block is found to be 14.4 J. The spring constant of the system (in N/m) is:
(1) 640 (2) 1280
(3) 2560 (4) 3200

Q34 The output Y of the logic circuit shown in figure will be:

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(1) $\left(\overline{A \cdot B}\right) + \overline{B}$
(2) $(A \cdot B) + \overline{B}$
(3) *A.B*
(4) *B*

Q35 A 50 g block of a metal at 150°C is dropped into an insulated calorimeter containing 100 g of water at 25°C. The final equilibrium temperature is 50°C. If the water equivalent of the calorimeter is 10 g, what is the specific heat capacity of the metal? (Specific heat of water = 1 cal/g°C)
(Assume no heat loss to surroundings)
(1) 0.44 cal/g°C
(2) 0.22 cal/g°C
(3) 0.55 cal/g°C
(4) 0.11 cal/g°C

Q36 Which of the following transition of electron for He^+ atom will have shortest wavelength of emitted photon? (symbols have usual meaning)
(1) $n = 3$ to $n = 2$
(2) $n = 15$ to $n = 4$
(3) $n = 2$ to $n = 1$
(4) $n = 15$ to $n = 2$

Q37 A string fixed at both ends oscillates in the first harmonic with a frequency of 480 Hz. If the amplitude at the middle of the string is 5.0 mm, what is the average speed of this point during one full oscillation?
(1) 4.8 m/s (2) 9.6 m/s
(3) 15.1 m/s (4) 19.2 m/s

Q38 Let I_0 be the intensity of polarised light after passing through the first polariser P_1 and it also passes through polarisers P_2 and P_3. If θ is the angle between transmission axes of P_1 and P_2 while $\left(\frac{\pi}{2} - \theta\right)$ is angle between transmission axes of P_2 and P_3, then transmitted intensity of emergent light from P_3 will be:
(1) $\frac{I_0}{4}\sin^2\left(2\theta\right)$
(2) $\frac{I_0}{4}\cos^2\left(2\theta\right)$
(3) $\frac{I_0}{2}\sin^2\left(2\theta\right)$
(4) $\frac{I_0}{2}\cos^2\left(2\theta\right)$

Q39 Two identical particles, each of mass *m*, move with the exact same speed *v* in opposite directions along two parallel straight lines separated by a distance *d*. If $\vec{L}_1$ is the total angular momentum of the system calculated about a point midway between the lines, and $\vec{L}_2$ is the total angular momentum calculated about a point sitting exactly on one of the lines, then:
(1) $\left|\vec{L}_1\right| = \frac{1}{2}\left|\vec{L}_2\right|$
(2) $\left|\vec{L}_1\right| = 2\left|\vec{L}_2\right|$
(3) $\left|\vec{L}_1\right| = \left|\vec{L}_2\right| = mvd$
(4) $\left|\vec{L}_1\right| = \left|\vec{L}_2\right| = 0$

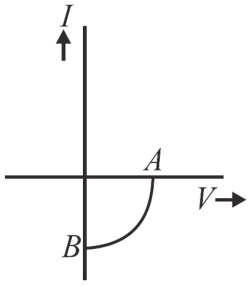
Q40 In Young's experiment, light of wavelength 4000Å is used to produce bright fringes of width 0.6 mm, at a distance of 2 m from the source. If the whole apparatus is dipped in a liquid of refractive index 1.5, then the fringe width will be:
(1) 0.2 mm (2) 0.3 mm
(3) 0.4 mm (4) 1.2 mm

Q41 Binding energy of a certain nucleus is 18×10^8 J. How much is the difference between total mass of all the nucleons and nuclear mass of the given nucleus?
(1) 20 μg (2) 2 μg
(3) 0.2 μg (4) 10 μg

Q42 Two coils *A* and *B* are connected in series across a 240 V, 50 Hz supply. The resistance of *A* is 5Ω and the inductance of *B* is 0.02 H. The power consumed is 3kW and the power factor is 0.75. The impedance of the circuit is;
(1) 0.144 Ω (2) 1.44 Ω
(3) 14.4 Ω (4) 144 Ω

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Q43 The given graph represents *V-I* characteristics for a semiconductor device.



Which of the following statements is **correct**?

(1) It is *V-I* characteristic for solar cell where, point *A* represents open circuit voltage and point *B* short circuit current.

(2) It is for a solar cell and points *A* and *B* represent open circuit voltage and open circuit current, respectively.

(3) It is for a photodiode and points *A* and *B* represent open circuit voltage and current, respectively.

(4) It is for a LED and points *A* and *B* represent open circuit voltage and short circuit current, respectively.

Q44 Considering rules of significant figures, the **correct** mathematical operation is:

(1) 1.30 + 9.132 = 10.4

(2) 17.388 × 1.0 = 17.388

(3) $\frac{15.50}{5.00} = 3.1$

(4) 3.24 – 1.2 = 2.0

Q45 If oscillatory electric field in a plane electromagnetic wave is given by $E = 0.6 \times 10^{-5} \sin(4000\,\pi x - 3 \times 10^{10}\,\pi t)$ T, then its wavelength is: (x is in metre and t is in second)

(1) 0.4×10^{-3} m

(2) 0.2×10^{-3} m

(3) 0.5×10^{-3} m

(4) 0.3×10^{-3} m

Q46 What formal charges are present in the compound $C_6H_5C \equiv N - O$?

(1) N is -1 and C is $+1$

(2) N is $+1$ and C is -1

(3) O is -1 and C is $+1$

(4) O is -1 and N is $+1$

Q47 Which of the following rate laws has an overall order of 0.5 for reaction involving substances x, y and z?

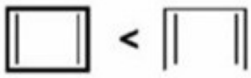
(1) $Rate = k(C_x)(C_y)(C_z)$

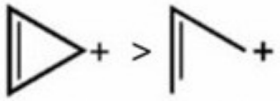
(2) $Rate = k(C_x)^{0.5}(C_y)^{0.5}(C_z)^{0.5}$

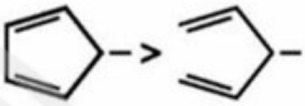
(3) $Rate = k(C_x)^{1.5}(C_y)^{-1}(C_z)^{0}$

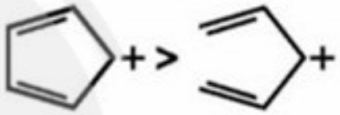
(4) $Rate = k(C_x)(C_z)^2/(C_y)^2$

Q48 Which of the following option is **incorrect** comparison of stability?

(1) 

(2) 

(3) 

(4) 

Q49 Solutions A, B, C and D are respectively 0.1 M glucose, 0.05 M NaCl, 0.05 M $BaCl_2$ and 0.1 M $AlCl_3$. Which one of the following pairs is isotonic?

(1) A and B (2) B and C

(3) A and D (4) A and C

Q50 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Pressure does not have any significant effect on solubility of solids in liquids.

Reason R: Pressure is intensive property.

In the light of the above statements, choose the correct answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

Q51

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The reaction in which hydrogen peroxide acts as a reducing agent is

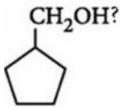
(1) $PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O$

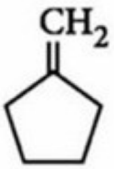
(2) $2KI + H_2O_2 \rightarrow 2KOH + I_2$

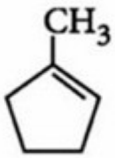
(3) $Ag_2O + H_2O_2 \rightarrow 2Ag + H_2O + O_2$

(4) $H_2SO_3 + H_2O_2 \rightarrow H_2SO_4 + H_2O$

Q52 Which is not a dehydration product of



(1) 

(2) 

(3) 

(4) 

Q53 What is the total spin value in case of $_{26}Fe^{3+}$ ion

(1) $+1$ or -1

(2) $+2$ or -2

(3) $+2.5$ or -2.5

(4) $+3$ or -3

Q54 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Tetrahedral complexes having unidentate ligands do not show geometrical isomerism.

Reason R: The relative position of unidentate ligands attached to central metal atom are same with respect to each other.

In the light of the above statements, choose the correct answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

Q55 Which of the following represents the **correct** order of their increasing bond order?

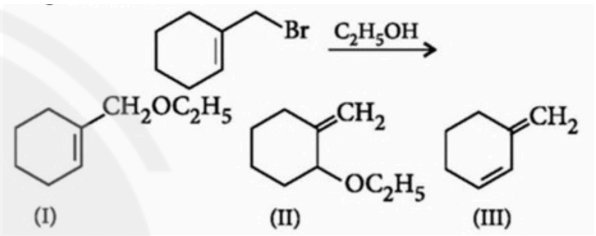
(1) $C_2^{2-} < He_2^+ < NO < O_2^-$

(2) $He_2^+ < O_2^- < NO < C_2^{2-}$

(3) $O_2^- < NO < C_2^{2-} < He_2^+$

(4) $NO < C_2^{2-} < O_2^- < He_2^+$

Q56 Which of the following products are possible in the given reaction?



(1) Only I (2) Only II

(3) Only III (4) All of these.

Q57 In which of the following molecules number of lone pairs and bond pairs on central atom are not equal?

(1) H_2O (2) I_3^-

(3) H_2S (4) SCl_2

Q58 If van't Hoff factors of aqueous solutions of X, Y and Z are 1.8, 0.8 and 2.5 then which of the following option is **correct**? (Assume equal concentrations in all three cases.)

(1) Elevation in boiling point : $X < Y < Z$

(2) Freezing point : $Z < X < Y$

(3) Osmotic pressure : $X = Y = Z$

(4) Vapour pressure : $Y < X < Z$

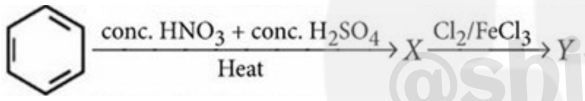
Q59 An aromatic amine (X) was treated with alcoholic potash and another compound (Y) then foul smelling gas C_6H_5NC is formed. The compound (Y) was formed by reacting compound (Z) with Cl_2 in the presence of slaked lime. The compound (Z) is

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(1) $CHCl_3$
(2) CH_3COCH_3
(3) CH_3OH
(4) $C_6H_5NH_2$

Q60 Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Magnetic moment of Mn^{2+} is higher than that of Cr^{2+}.
Reason R: Higher the atomic number smaller is the magnetic moment.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

Q61 Identify Y in the following sequence of reaction.

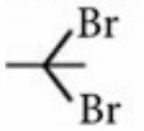


(1) p-chloronitrobenzene
(2) m-chloronitrobenzene
(3) o-chloronitrobenzene
(4) o,p-dichloronitrobenzene

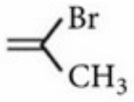
Q62 Which of the following is present as five membered cyclic structure (furanose structure)?
(1) Ribose (2) Glucose
(3) Maltose (4) Galactose

Q63 Which of the following increasing order of oxidising power is **correct** for the following species? $VO_2^+, MnO_4^-, Cr_2O_7^{2-}$
(1) $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$
(2) $VO_2^+ < MnO_4^- < Cr_2O_7^{2-}$
(3) $Cr_2O_7^{2-} < VO_2^+ < MnO_4^-$
(4) $Cr_2O_7^{2-} < MnO_4^- < VO_2^+$

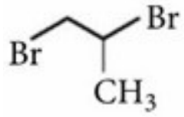
Q64 1-Methyl ethylene oxide when treated with an excess of HBr produces
(1)



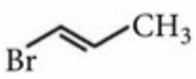
(2)



(3)



(4)



Q65 Given below are two statements:
Statement I: Carbon tetrachloride depletes the ozone layer.
Statement II: Chloroform is used in the production of freon refrigerant R-22.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

Q66 The IUPAC name of the complex $[Pt(NH_3)_2Cl(NH_2CH_3)]Cl$ is:
(1) diammine(methanamine)chloridoplatinum(II) chloride
(2) diamminechlorido(methanamine)platinum(II) chloride
(3) diamminechlorido(aminomethane)platinum(II) chloride
(4) bisammine(methanamine)chloridoplatinum(II) chloride.

Q67 Given below are two statements:
Statement I: Fullerenes have dangling bonds.
Statement II: Hybridisation of carbon in diamond is sp^2.
In the light of the above statements, choose the most appropriate answer from the options given

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below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

Q68 If pK_a of monobasic acid HA is 4 then its van't Hoff factor of $0.01\ \mathrm{M}$ solution is
(1) 1.10 (2) 1.01
(3) 1.20 (4) 1.02

Q69 Match the **List-I** with **List-II**.

List-I (Oxide)		List-II (Example)	
(A)	Neutral oxide	(I)	SO_3
(B)	Acidic oxide	(II)	As_2O_3
(C)	Basic oxide	(III)	MgO
(D)	Amphoteric oxide	(IV)	CO

Choose the **correct** answer from the options given below:
(1) A-IV; B-I, C-III; D-II
(2) A-II; B-III, C-I; D-IV
(3) A-III; B-IV, C-II; D-I
(4) A-IV; B-II, C-I; D-III

Q70 Consider the following sequence of reactions :

$$Glucose \xrightarrow[\text{Excess}]{C_6H_5NHNH_2} X \xrightarrow[\Delta]{\text{dil.HCl}} Y \xrightarrow{Zn/CH_3COOH} Z$$

The final product (Z) is
(1) sorbitol (2) fructose
(3) mannose (4) mannitol

Q71 Identify 'X' in the following sequence of reactions :

$$\underset{(X)}{C_4H_7OCl} \xrightarrow{NH_3} C_4H_9ON \xrightarrow{Br_2/KOH} CH_3CH_2CH_2NH_2$$

(1) $(CH_3)_2CHCOCl$
(2) $CH_3CH_2CH(OH)CH_2Cl$
(3) $CH_3CH_2CH_2COCl$
(4) $ClCH_2CH_2CH_2CHO$

Q72 Predict the product (C) for the following :

$$C_3H_6 \xrightarrow{Br_2} (A) \xrightarrow[\text{(ii) } NaNH_2/\text{Liq. } NH_3]{\text{(i) alc. KOH}} (B) \xrightarrow{Hg^{2+}/H_2SO_4} (C)$$

(1) $CH_3CH{=}CH_2$
(2) $CH_3{-}\underset{\displaystyle OH}{\underset{|}{CH}}{-}CH_3$
(3) $CH_3{-}\underset{\displaystyle OH}{\underset{|}{CH}}{-}\underset{\displaystyle Br}{\underset{|}{CH_2}}$
(4) $CH_3{-}\overset{\displaystyle O}{\overset{\|}{C}}{-}CH_3$

Q73 If the bond energies of $H-H$, $Br-Br$, and $H-Br$ are 433, 192 and 364 kJ mol^{-1} respectively, the ΔH° for the reaction, $H_{2(g)} + Br_{2(g)} \rightarrow 2HBr_{(g)}$ is
(1) -261 kJ mol^{-1}
(2) $+103$ kJ mol^{-1}
(3) $+261$ kJ mol^{-1}
(4) -103 kJ mol^{-1}

Q74 Match **List-I** with **List-II.**

List-I (Metal ion)		List-II (Group in qualitative analysis)	
(A)	Mn^{2+}	(I)	Group-III
(B)	As^{3+}	(II)	Group-II A
(C)	Cu^{2+}	(III)	Group-IV
(D)	Al^{3+}	(IV)	Group-II B

Choose the **correct** answer from the options given below:
(1) A-III; B-I, C-II; D-IV
(2) A-II; B-III, C-I; D-IV
(3) A-III; B-IV, C-II; D-I
(4) A-IV; B-II, C-I; D-III

Q75 In a zero-order reaction for every $10^\circ C$ rise of temperature, the rate is doubled. If the temperature is increased from $10^\circ C$ to $100^\circ C$, the rate of the reaction will become
(1) 256 times (2) 512 times
(3) 64 times (4) 128 times.

Q76 Match **List-I** with **List-II.**

List-I	List-II

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(A)	Second order reaction	(I)	Natural radioactive decay
(B)	Zero order reaction	(II)	Unit of rate constant is $M^{-1}s^{-1}$
(C)	First order reaction	(III)	Inversion of cane sugar
(D)	Pseudo first order reaction	(IV)	Thermal decomposition of HI on gold surface

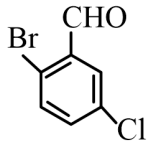
Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-IV, D-I
(2) A-II, B-I, C-IV, D-III
(3) A-II, B-IV, C-I, D-III
(4) A-I, B-III, C-IV, D-II

Q77 For vaporization of water at 1 atmospheric pressure, the values of ΔH and ΔS are $40.63\ \text{kJ mol}^{-1}$ and $108.8\ \text{JK}^{-1}\ \text{mol}^{-1}$, respectively. The temperature when Gibbs' energy change (ΔG) for this transformation will be zero, is

(1) $273.4\ \text{K}$
(2) $393.4\ \text{K}$
(3) $373.4\ \text{K}$
(4) $293.4\ \text{K}$

Q78 The correct IUPAC name of the given compound



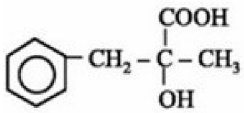
is

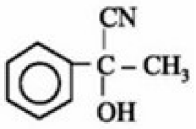
(1) *o*-bromo-*p*-chlorobenzaldehyde
(2) 2-bromo-5-chlorobenzaldehyde
(3) 6-bromo-3-chlorobenzaldehyde
(4) 1-bromo-4-chlorobenzaldehyde.

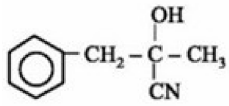
Q79 In the following reactions acetic acid yielded a product D.

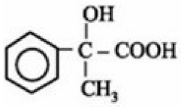
$$CH_3COOH \xrightarrow{SOCl_2} A \xrightarrow[\text{anhy.}AlCl_3]{\text{benzene}} B \xrightarrow{HCN} C \xrightarrow{H_2O} D$$

The structure of D would be

(1) 

(2) 

(3) 

(4) 

Q80 Match **List-I** with **List-II.**

List-I (Reagent)		**List-II (Process)**	
(A)	$NaOH + CaO$	(I)	Decarboxylation of sodium salts of carboxylic acids
(B)	$V_2O_5 + Al_2O_3$	(II)	Isomerisation of alkanes
(C)	anhyd. $AlCl_3 + HCl$	(III)	Hydrogenation of alkynes
(D)	$H_2 + Pd/C$	(IV)	Dehydrogenation of alkanes

Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-IV, D-I
(2) A-I, B-III, C-II, D-IV
(3) A-I, B-IV, C-II, D-III
(4) A-IV, B-III, C-II, D-I

Q81 Complete combustion of a hydrocarbon gives $0.66\ \text{g}$ of CO_2 and $0.36\ \text{g}$ of H_2O. Then the empirical formula of the compound is

(1) C_3H_8
(2) C_2H_2
(3) C_5H_{10}
(4) CH_4

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Q82 Benzamide can be converted to benzonitrile with
(1) H_3O^+
(2) OH^-/H_2O
(3) KCN
(4) P_2O_5, heat.

Q83 Given below are two statements:
Statement I: Mole fraction is temperature independent concentration term.
Statement II: Sugar dissolves readily in benzene.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

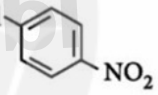
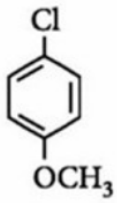
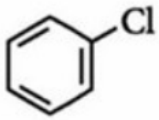
Q84 The number of orbitals associated with quantum numbers $n = 5, m_s = +\frac{1}{2}$ is
(1) 25 (2) 11
(3) 15 (4) 50

Q85 Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The two strands of DNA are complementary to each other.
Reason R: Adenine specifically forms hydrogen bonds with guanine whereas cytosine forms hydrogen bonds with thymine.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

Q86 Given that the standard potentials (E°) of Cu^{2+}/Cu and Cu^+/Cu are 0.34 V and 0.522 V respectively, the E° of Cu^{2+}/Cu^+ is
(1) -0.182 V
(2) $+0.158$ V
(3) -0.158 V
(4) 0.182 V.

Q87 In the Carius method of estimation (a gravimetric determination of phosphorus, 0.248 g of an organic compound gave a precipitate of $Mg_2NH_4PO_4$ which on ignition yielded 0.444 g of $Mg_2P_2O_7$. What is the percentage of phosphorus in the compound?
(1) 68% (2) 50%
(3) 42% (4) 78%

Q88 The dipole moments of $CCl_4, CHCl_3$ and CH_4 are in the order
(1) $CH_4 = CCl_4 < CHCl_3$
(2) $CHCl_3 < CH_4 = CCl_4$
(3) $CH_4 < CCl_4 < CHCl_3$
(4) $CCl_4 < CH_4 < CHCl_3$

Q89 Which of the following compounds undergoes nucleophilic substitution reaction most easily?
(1) 
(2) 
(3) 
(4) 

Q90 Match **List-I** with **List-II.**

List-I (Vitamin)	List-II (deficiency disease)

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(A)	Riboflavin	(I)	Beriberi
(B)	Thiamine	(II)	Scurvy
(C)	Pyridoxine	(III)	Cheilosis
(D)	Ascorbic acid	(IV)	Convulsions

Choose the **correct** answer from the options given below:

(1) A-I, B-IV, C-III, D-II
(2) A-III, B-I, C-IV, D-II
(3) A-III, B-IV, C-I, D-II
(4) A-IV, B-II, C-I, D-III

Q91 Match **List-I** with **List-II**.

List-I		List-II	
(A)	Har Gobind Khorana	(I)	Developed a system of enzymatic synthesis of RNA
(B)	George Gamow	(II)	First to elucidate a transcriptionally regulated system
(C)	Severo Ochoa	(III)	Developed chemical method for synthesis of RNA homopolymers and copolymers
(D)	Jacob and Monod	(IV)	Argued that genetic the code should constitute a combination of bases

Choose the most appropriate answer from the options given below.

(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-I, C-II, D-III
(3) A-III, B-II, C-I, D-IV
(4) A-II, B-IV, C-I, D-III

Q92 A newly discovered multicellular organism has stored food as floridean starch. Identify the **correct** statement about it.

(1) The major photosynthetic pigment in it is fucoxanthin.
(2) It is most likely found in terrestrial habitat.
(3) Its gametes are non-motile.
(4) Its asexual spores have laterally attached flagella.

Q93 Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): If the F_2 phenotypic ratio of a monohybrid cross deviates from 3:1, this indicates a case of complete dominance of an allele over the other one.

Reason (R): In Mendelian monohybrid cross, the recessive allele did not express in F_1 generation.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

Q94 Identify the **correct** statements regarding the characteristics of a prokaryotic cells.

A. Many of them possess a tightly bound three-layered cell envelope.
B. The genetic material is absent.
C. Due to lack of membrane-bound organelles, ATP is not synthesised in them.
D. Ribosomes are the only organelles found, present in the cytoplasm.
E. They may or may not possess flagella.

Choose the **correct** option.

(1) A, D and E (2) B and E only
(3) C and D only (4) A, B and D

Q95 In the process of decomposition:

(1) prokaryotic organisms perform the fragmentation of detritus.
(2) low temperature and anaerobic conditions lead to its reduced rate.
(3) nitrogen and water-soluble substances of detritus are catabolised at a very low rate.
(4) only eukaryotic organisms perform catabolism of organic molecules.

Q96 In papaya:

(1) flowers, are hermaphrodite.

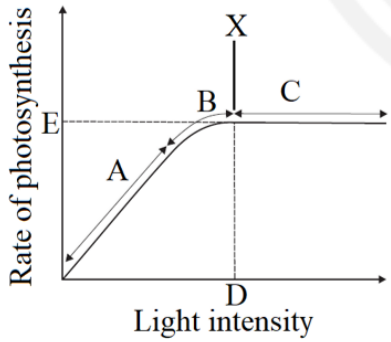
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(2) autogamy is the only mode of pollination.
(3) inbreeding depression is not likely to occur.
(4) geitonogamy is prevented because pollen release and stigma receptivity are not synchronised.

Q97 An ecologist observing a population of beetles growing in a habitat with unlimited resources would most likely see which pattern of growth?
(1) Sigmoid curve, described by the equation dN/dt = rN(K-N/K)
(2) J-shaped curve, described by the equation dN/dt = rN
(3) A stable population with zero growth
(4) A logistic growth pattern with a long lag phase

Q98 Select the **correctly** matched pair from the following.
(1) Slime moulds - Saprophytic protists
(2) Diatoms - Cause red tides
(3) Euglenoids - Stiff cellulose plates on outer surface
(4) Protozoans - Chief producers in oceans

Q99 Shown below is a graph demonstrating effect of light intensity on the rate of photosynthesis. Identify the **incorrect** statement regarding it.



(1) A and B represent increase in rate of photosynthesis due to increase in light intensity.
(2) X represents light saturation i.e. 10% of full sunlight.
(3) Beyond X, increase in light intensity has no effect on rate of photosynthesis.
(4) At C, light becomes the limiting factor for plants in shade or in dense forests.

Q100 A student observes a transverse section of a root and notes the following features: epidermis, cortex, endodermis, pericycle, large and well-developed pith, and more than six xylem bundles. The root is most likely from a:
(1) Sunflower plant.
(2) Dicotyledonous plant.
(3) Monocotyledonous plant.
(4) Gymnosperm.

Q101 Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Mitosis is called equational division.
Reason (R): After a complete mitotic cell cycle, the amount of DNA in parent and daughter cell will be same.
In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

Q102 In a biogas plant, the production of biogas occurs due to the activity of a type of:
(1) true bacteria.
(2) archaebacteria.
(3) imperfect fungi.
(4) cyanobacteria.

Q103 Match **List-I** with **List-II**.

	List-I (Process)		**List-II (Net Gain)**
(A)	One turn of EMP pathway	(I)	1 NADH + H^+
(B)	One turn of TCA cycle	(II)	2 ATP, 2 Pyruvate, 2 NADH + H^+
(C)	Conversion of one pyruvate into acetyl	(III)	1 ATP, 3 NADH + H^+, 1 $FADH_2$

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	CoA		
(D)	Transport of 12 protons across F_0 particle of ATP synthase	(IV)	3 ATP

Choose the most appropriate answer from the options given below.
(1) A-II, B-III, C-I, D-IV
(2) A-II, B-I, C-III, D-IV
(3) A-III, B-II, C-I, D-IV
(4) A-I, B-III, C-II, D-IV

Q104 In the *lac* operon, what is the role of lactose?
(1) It acts as a repressor, binding to the operator and preventing transcription.
(2) It acts as an inducer, inactivating the repressor protein.
(3) It binds to the promoter region, enhancing RNA polymerase activity.
(4) It is a co-repressor that activates the repressor protein.

Q105 Match **List-I** with **List-II**.

List-I (Genetic Disorder)		**List-II (Characteristic)**	
(A)	Sickle-cell anaemia	(I)	Individual has 47 chromosomes
(B)	Phenylketonuria	(II)	Mutation of GAG to GUG
(C)	Down's Syndrome	(III)	Individual has 45 chromosomes
(D)	Turner's Syndrome	(IV)	Associated gene is pleiotropic

Choose the most appropriate answer from the options given below.
(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-I, C-IV, D-II
(4) A-II, B-IV, C-I, D-III

Q106 Understanding the structural basis of genotype and phenotype conversion became the focus of attention after:
(1) The work of Mendel and others who followed him.
(2) The discovery of the mechanism of evolution.
(3) Contributions from Watson, Crick, Nirenberg, and Khorana.
(4) The tackling of parallel problems in structural biology.

Q107 Which of the following statements about ecological pyramids are correct?
A. The base of each pyramid represents the producers.
B. The pyramid of energy is always upright.
C. The pyramid of biomass in the sea is generally inverted.
D. A given species, that occupies more than one trophic level in the same ecosystem, is placed in all concerned levels of the pyramid.
E. Saprophytes are given a prominent place at the base of ecological pyramids.
Choose the **correct** option.
(1) A, B, C only
(2) A, B, D, E only
(3) A, B, C, D only
(4) B, C, D only

Q108 Which of the following regions has the highest number of bird species?
(1) New York (2) India
(3) Greenland (4) Colombia

Q109 A farmer wants to induce rooting in stem cuttings of a rose plant and also wants to prepare weed-free lawns. Which plant growth regulator would be most suitable for both these purposes?
(1) Gibberellin (2) Cytokinin
(3) Auxin (4) Ethylene

Q110 Given below are two statements:
Statement I: The meristem that occurs in the mature regions of roots and shoots and is responsible for producing secondary tissues is called the lateral meristem.
Statement II: Fascicular vascular cambium, interfascicular cambium and cork-cambium are examples of primary meristems.
In the light of the above statements, choose the correct answer from the options given below:

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(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

Q111 Dog and Gorilla share common:
A. Order
B. Family
C. Kingdom
D. Class
E. Phylum
(1) A, B, C, D (2) C, D, E
(3) B, C, D (4) A, C, E

Q112 Identify the **incorrect** statement about axillary bud.
(1) They develop into tendrils for climbing in *Citrus*.
(2) They develop into thorns for protection in *Bougainvillea*.
(3) Usually they develop into a branch.
(4) They are not present in the axil of leaflets.

Q113 Match **List-I** with **List-II**.

List-I (Cellular Structure/Organelle)		**List-II (Characteristic)**	
(A)	Cilia	(I)	Absorption of digested food materials
(B)	Microvilli	(II)	Possesses hydrolytic enzymes that digest biomolecules
(C)	Microbodies	(III)	Meant for locomotion
(D)	Lysosome	(IV)	Present in both plant and animal cells

Choose the most appropriate answer from the options given below.
(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-I, C-IV, D-II
(4) A-II, B-IV, C-I, D-III

Q114 Read the following statements regarding members of Kingdom Fungi.
A. All of them are heterotrophic.
B. None of them is filamentous.
C. Cell wall may or may not be present.
D. Those that depend on living plants and animals are called saprophytes.
E. The network of hyphae is known as mycelium.
In the light of above statements choose the **correct** answer from the options given below:
(1) A, C and E are correct, B and D are incorrect
(2) B, D and E are correct, A and C are incorrect
(3) A is correct, B, C, D and E are incorrect
(4) A and E are correct, B, C and D are incorrect

Q115 Arrange the given steps of transcription in bacteria in **correct** sequence.
A. Binding of rho factor to RNA polymerase.
B. Elongation of the polynucleotide chain.
C. Falling off of the nascent RNA.
D. Binding of sigma factor to RNA polymerase.
Choose the **correct** option:
(1) D → B → A → C
(2) A → B → C → D
(3) C → A → B → D
(4) B → C → A → D

Q116 The probe that is used in DNA fingerprinting:
(1) is a type of polypeptide.
(2) has a size of about 100 to 200 kb.
(3) is radiolabelled for being used.
(4) lacks repetitive sequences.

Q117 Given below are two statements:
Statement I: Pollination in corn is a chance factor.
Statement II: Pollination in *Vallisneria* is a chance factor.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.

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(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

Q118 Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): The sex organs in bryophytes are borne on gametophytes.
Reason (R): The gametes in bryophytes are produced by meiosis in cells of sex organs.
In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

Q119 Given below are two statements:
Statement I: Chemiosmosis is primarily required for synthesis of ATP and NADH + H^+ in chloroplasts.
Statement II: The enzyme FNR involved in chemiosmosis is a type of reductase.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

Q120 Read the following statements and find the suitable option.
A. Generally petals are green, leaf like and protect the flower in the bud stage.
B. Sepals are brightly coloured to attract insects for pollination.
C. If stamens in a flower remain free this is called polyandrous.
D. Each ovary bears one or more ovules attached to a flattened, cushion like placenta.
E. There may be a variation in the length of filaments of stamen within a flower, as in *Salvinia* and mustard.
(1) A, B and D are true but C and E are false
(2) C and D are true but A, B and E are false
(3) D and E are true but A, B and C are false
(4) C, D and E are true but A and B are false

Q121 What happens during Anaphase I of meiosis?
(1) Sister chromatids separate.
(2) Homologous chromosomes separate, while sister chromatids remain associated.
(3) The centromere of each chromosome splits.
(4) The nuclear membrane and nucleolus reappear.

Q122 Identify the **incorrectly** matched pair.
(1) *Saccharomyces cerevisiae* - Reproduces asexually by budding
(2) *Saccharomyces cerevisiae* - Used for making bread
(3) *Saccharomyces cerevisiae* - Used for making beer
(4) *Saccharomyces cerevisiae* - Used to obtain clot buster enzyme

Q123 The physico-chemical approach to study and understand living organisms, where concepts from physics and chemistry are applied in studies, provides information about all the following, **except**;
(1) Various processes in molecular terms.
(2) What types of organic compounds are present in living organisms.
(3) Role of organic compounds in a dead cell.
(4) Explain the abnormal processes that occur during any diseased condition.

Q124 Match **List-I** with **List-II**.

List-I		List-II	
(A)	Ferns	(I)	Gymnosperms

(B)	Mosses	(II)	Angiosperms
(C)	Naked seeds	(III)	Pteridophytes
(D)	Fruits	(IV)	Bryophytes

Choose the most appropriate answer from the options given below.
(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-I, C-IV, D-II
(4) A-II, B-IV, C-I, D-III

Q125 Consider the following characteristics regarding vascular bundles in plants:
A. Xylem and phloem are arranged in an alternate manner on different radii.
B. Xylem and phloem are situated along the same radius.
C. Cambium is present between xylem and phloem.
D. Cambium is not present between xylem and phloem.
Which combination of characteristics describes a conjoint, open vascular bundle?
(1) A and C (2) B and C
(3) A and D (4) B and D

Q126 Identify the **incorrect** statement about respiration.
(1) The respiratory pathway is traditionally considered as anabolic pathway.
(2) Fatty acids are first degraded to acetyl CoA before entering the respiratory pathway.
(3) When the organism needs to synthesise fatty acids, acetyl CoA is withdrawn from the respiratory pathway.
(4) Because the respiratory pathway is involved in both anabolism and catabolism, it is considered an amphibolic pathway.

Q127 Given below are two statements:
Statement I: DNA and RNA both can undergo mutation.
Statement II: Uracil is a type of purine.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

Q128 The incorrect feature of grass family is:
(1) perianth present.
(2) inferior ovary.
(3) single seed in fruit.
(4) presence of scutellum in embryo.

Q129 The infectious agent that causes potato spindle tuber disease:
(1) has abnormally folded protein as the infectious component.
(2) cannot pass through bacteria-proof filters.
(3) is considered to be non-cellular.
(4) doesn't possess either DNA or RNA.

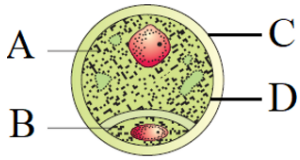
Q130 In a dihybrid cross conducted by Mendel between pea plants with round yellow seeds (RRYY) and wrinkled green seeds (rryy), the F_2 generation showed a phenotypic ratio of 9 : 3 : 3 : 1. This observation led to the formulation of:
(1) Law of Dominance.
(2) Law of Segregation.
(3) Law of Independent Assortment.
(4) Principle of Co-dominance.

Q131 The microtubules are not a component of:
(1) axoneme of flagellum.
(2) kinetochore of a chromosome.
(3) cytoskeleton in a eukaryotic cell.
(4) mitotic apparatus.

Q132 During nutrient cycling in an ecosystem:
(1) nutrients are repeatedly used.
(2) nutrients cannot be stored by any component of the ecosystem.
(3) a structural feature of the ecosystem is studied.
(4) only gaseous nutrients can be recycled.

Q133 Go through the mature pollen grain shown below and choose the **correct** statement regarding it.

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(1) A is made of sporopollenin.
(2) B gives rise to male gametes by mitosis.
(3) C is a continuous wall layer.
(4) D has components that cannot be degraded by any known enzyme.

Q134 The intercellular factor that controls development in plants is:
(1) PGR.
(2) genetic factor.
(3) nutrients.
(4) oxygen.

Q135 Identify the **incorrect** statement.
(1) Reserpine is produced by *Calotropis*.
(2) Strychnine is produced by plants for defence against grazers and browsers.
(3) India has more than 50,000 genetically different strains of rice.
(4) India has more than 1,000 varieties of mango.

Q136 Which specific part of the neuron contains typical cell organelles along with certain granular bodies known as Nissl's granules?
(1) Axon (2) Myelin sheath
(3) Cell body (4) Synaptic knob

Q137 Analyse the following statements comparing Chordates and Non-chordates.
(A) In chordates, the central nervous system is dorsal, hollow, and single.
(B) In non-chordates, the central nervous system is ventral, solid, and double.
(C) Chordates have a ventral heart, whereas non-chordates have a dorsal heart.
(D) Pharyngeal gill slits are absent in chordates but present in non-chordates.
(E) A post-anal tail is a characteristic presence in chordates.
Identify the **correct** statements:
(1) A, B, C and E (2) A, B, D and E
(3) B, C, D and E (4) A, C, D and E

Q138 What is the total number of walking legs present on a cockroach?
(1) 3 (2) 4
(3) 6 (4) 8

Q139 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The most distinctive feature of echinoderms is the presence of a water vascular system.
Reason R: This system takes over the entire role of the absent excretory system by actively filtering blood plasma and removing nitrogenous wastes.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

Q140 How many of the following given in the box below are not of clinical diagnostic significance?

Heart sounds, Blood pressure, Pulse rate, ECG

Choose the **correct** option.
(1) Two (2) Zero
(3) Four (4) One

Q141 Which of the following changes is usually observed when a person goes on a trip to mountains from plains?
(1) Increase in production of red blood cells
(2) Decrease in breathing rate to save energy
(3) Increase breakdown of platelets
(4) Increase breakdown of erythrocytes

Q142 All of the following help in maintaining the pH and ionic balance of the body fluids by selective secretion of hydrogen ions, **except:**
(1) PCT
(2) Bowman's capsule
(3) DCT
(4) Collecting duct

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Q143 Match **List-I** with **List-II** w.r.t frog.

List-I		List-II	
(A)	Liver	(I)	Absorbs the final digested food
(B)	Gall bladder	(II)	Produces digestive enzymes
(C)	Pancreas	(III)	Secretes bile
(D)	Villi and Microvilli	(IV)	Temporarily stores bile

Choose the **correct** answer from the options given below.

(1) A-III, B-IV, C-II, D-I
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-I, C-II, D-IV
(4) A-II, B-IV, C-III, D-I

Q144 During agarose gel electrophoresis, DNA fragments separate according to their size as they move toward the anode. Which specific inherent property of the agarose matrix enables this physical separation?

(1) Ion-exchange capability
(2) Enzymatic degradation
(3) Radioactive tagging
(4) Sieving effect

Q145 Which phylum contains organisms that exhibit an organ-system level of body organization, bilateral symmetry, and characteristic segmentation forming distinct external and internal rings?

(1) Arthropoda
(2) Mollusca
(3) Aschelminthes
(4) Annelida

Q146 Plasma constitutes nearly 'X' per cent of the blood and 'Y' per cent of blood plasma is water. Select the **correct** option for 'X' and 'Y' respectively w.r t. humans.

(1) 45 and 90-92
(2) 55 and 90-92
(3) 45 and 6-8
(4) 55 and 6-8

Q147 Transgenic mice are extensively developed for use in:

(1) Mass producing alpha-1-antitrypsin for emphysema.
(2) Testing the safety of batches of vaccines, such as the polio vaccine.
(3) Producing human insulin on an industrial scale.
(4) Extracting human growth hormone.

Q148 Given below are two statements.

Statement I: In humans, each kidneys have nearly one million complex tubular structures called nephrons.

Statement II: Blood from the glomerulus is carried away by an efferent arteriole.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Q149 Which specific class of enzymes uniquely catalyses the linking together of two compounds, such as the joining of C-O, C-S, C-N, or P-O bonds?

(1) Transferases
(2) Lyases
(3) Hydrolases
(4) Ligases

Q150 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Recombinant therapeutics generally do not induce unwanted immunological responses in human patients, unlike similar products isolated from non-human animal sources.

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Reason R: Human insulin produced synthetically via recombinant DNA technology in bacteria is structurally and chemically absolutely identical to the natural human insulin molecule.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

Q151 Which of the following is a hypothalamic inhibiting hormone?

(1) Gonadotrophin releasing hormone (GnRH)
(2) Somatostatin
(3) Prolactin
(4) Oxytocin

Q152 Match **List-I** with **List-II**.

List-I		List-II	
(A)	Condoms	(I)	Prevent the physical meeting of ovum and sperms
(B)	Copper T (CuT)	(II)	Suppress sperm motility and lower fertilizing capacity
(C)	Oral pills	(III)	Inhibit ovulation and implantation while altering cervical mucus
(D)	Vasectomy	(IV)	Block gamete transport entirely through surgical intervention

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-II, B-IV, C-I, D-III
(4) A-I, B-II, C-III, D-IV

Q153 A population of finch birds with medium sized beaks colonizes an island with plants producing medium and large sized seeds. After a disease wipes out all the plants with medium sized seeds, what is most likely to happen to the finches?

(1) They will become extinct.
(2) Their beaks will not change at all.
(3) Their beaks become smaller.
(4) Their beaks will become larger.

Q154 Read the following statements regarding the frog.

(A) The frog's heart is three-chambered.
(B) In a male frog, the ureters act strictly as excretory ducts.
(C) Frogs possess a well-developed closed circulatory system but completely lack a lymphatic system.
(D) Partially digested food passed from the stomach into the duodenum is called chyme.
(E) Frogs lack external ears.

Which of the above statements are **correct**?

(1) B, C, and D
(2) A, B, and C
(3) A, D, and E
(4) C, D, and E

Q155 Common cold is mostly caused by Rhinoviruses. Which of the following treatments will provide the best cure from this infectious agent?

(1) A glass of Oral Rehydration Suspension (ORS) plus two tablets of streptomycin twice a day for three days.
(2) Two tablets of amoxycillin twice a day for at least three days.
(3) One tablet of amoxycillin plus one tablet of a pain killer four times a day for at least three days.
(4) Adequate rest and a balanced diet for at least three days.

Q156 Given below are two statements.

Statement I: Cell-mediated immunity (CMI) is driven by T-lymphocytes which is responsible for graft rejection.

Statement II: The humoral immune response involves B-lymphocytes synthesizing an army of

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polysaccharides called antibodies.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

Q157 Which two respiratory volumes sum up to form Expiratory Capacity?

(1) Tidal Volume (TV) + Inspiratory Reserve Volume (IRV)

(2) Expiratory Reserve Volume (ERV) + Residual Volume (RV)

(3) Tidal Volume (TV) + Expiratory Reserve Volume (ERV)

(4) Tidal Volume (TV) + Residual Volume (RV)

Q158 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Plasmids and bacteriophages are extensively used as cloning vectors in recombinant DNA technology.

Reason R: They possess the inherent ability to replicate within bacterial cells independent of the control of chromosomal DNA.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is not the correct explanation of A.

Q159 All of the following can increase the rate of heart beat, **except**

(1) Sympathetic neural signals

(2) Epinephrine

(3) Parasympathetic neural signals

(4) Nor-adrenaline

Q160 Identify the **incorrectly** matched pair.

(1) Protonephridia - Platyhelminthes (e.g., *Planaria*)

(2) Nephridia - Annelids (e.g., Earthworms)

(3) Malpighian tubules - Cephalochordates (e.g., *Amphioxus*)

(4) Antennal glands (Green glands) - Crustaceans (e.g., prawns)

Q161 Between which two distinct anatomical organs do the hepatic portal vein directly carry blood?

(1) From the liver directly to the heart

(2) From the intestine directly to the liver

(3) From the kidneys directly to the liver

(4) From the stomach directly to the heart

Q162 Match **List-I** with **List-II.**

List-I		**List-II**	
(A)	Trophoblast	(I)	An early embryo consisting of 8 to 16 blastomeres
(B)	Inner cell mass	(II)	An inner group of cells attached to the trophoblast that differentiates into the embryo proper
(C)	Chorionic villi	(III)	The outer layer of the blastocyst that attaches to the endometrium
(D)	Morula	(IV)	Finger-like projections appearing after implantation that interdigitate with uterine tissue

Choose the **correct** answer from the options given below:

(1) A-IV, B-II, C-III, D-I

(2) A-III, B-I, C-IV, D-II

(3) A-II, B-III, C-I, D-IV

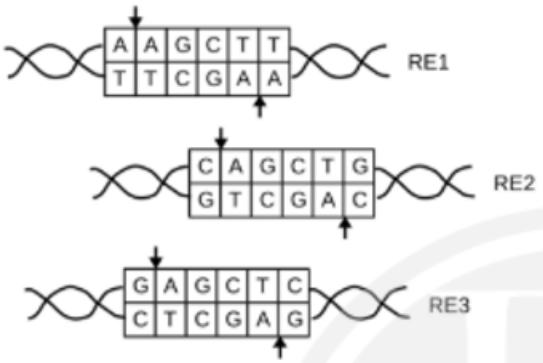
(4) A-III, B-II, C-IV, D-I

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Q163 Which among the following is **not** the secretion of sebaceous glands?

(1) Sterols (2) Hydrocarbons
(3) Lactic acid (4) Waxes

Q164 Provided below are recognition sequences for restriction enzymes 1 (RE1), 2 (RE2) and 3 (RE3). Arrows indicate the positions where the enzymes digest on the two strands. Which of the following can the RE1 digested DNA ligate to?



(1) All three RE1, RE2 and RE3 digested DNA.
(2) Only to RE1 digested DNA.
(3) Only to RE1 and RE3 digested DNA.
(4) Only to RE1 and RE2 digested DNA.

Q165 Select the **incorrect** statement from the following.

(1) TCT and PTH together regulate blood calcium levels.
(2) Hypersecretion of hormones by the adrenal cortex leads to Addison's disease.
(3) Catecholamines stimulate glycogenolysis resulting in hyperglycaemia.
(4) Estrogen regulates female sexual behaviour.

Q166 Traditional breeding techniques often lead to the inclusion of undesirable genes along with the desired ones. Recombinant DNA technology overcomes this limitation by:

(1) Inducing targeted mutations exclusively in the undesirable genes.
(2) Isolating and introducing only one or a specific set of desirable genes into the target organism.
(3) Fusing entire genomes of two organisms and selectively silencing undesirable genes via RNAi.
(4) Utilizing physical mutagens to destroy unwanted DNA sequences prior to breeding.

Q167 Given below are two statements.

Statement I: Ernst Haeckel proposed embryological support for evolution based on the observation of certain rudimentary features during embryonic stages common to all vertebrates.

Statement II: Karl Ernst von Baer enthusiastically supported Haeckel's proposal, proving conclusively that embryos always pass through the adult stages of other ancient animals.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

Q168 A skeletal muscle fiber is described as a syncytium because:

(1) It contains a large number of parallelly arranged myofilaments.
(2) Its sarcoplasm contains many nuclei.
(3) It is innervated by multiple motor neurons.
(4) It is highly rich in myoglobin.

Q169 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Inside the mature Graafian follicle, the secondary oocyte forms a new acellular membrane called the zona pellucida surrounding it.

Reason R: During fertilisation, a sperm comes in contact with the zona pellucida layer of the ovum and induces changes in the membrane that block the entry of additional sperms.

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In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

Q170 Match **List-I** with **List-II**.

List-I		List-II	
(A)	F-actin	(I)	Masks the active binding sites for myosin.
(B)	Tropomyosin	(II)	Polymer of 'G' (globular) actins.
(C)	Troponin	(III)	Runs close to the F-actins throughout their length.
(D)	Heavy Meromyosin (HMM)	(IV)	Globular head with a short arm.

Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-III, D-IV
(2) A-II, B-III, C-I, D-IV
(3) A-I, B-II, C-III, D-IV
(4) A-II, B-IV, C-I, D-III

Q171 Which of the following **accurately** describes the anatomy of the scapula?
(1) A large triangular flat bone situated dorsally between the 2nd and 7th ribs.
(2) A long slender bone with two curvatures situated ventrally.
(3) A cup-shaped bone articulating with the humerus and radius.
(4) A fused bone comprising the ilium, ischium, and pubis.

Q172 Identify the **incorrectly** matched pair.
(1) *Plasmodium falciparum* - Malignant malaria
(2) *Wuchereria bancrofti* - Elephantiasis
(3) *Haemophilus influenzae* - Common cold
(4) *Microsporum* – Ringworms

Q173 What was the main biological challenge Eli Lilly faced when trying to produce human insulin using recombinant DNA technology?
(1) *E. coli* cells rapidly degraded the human insulin gene.
(2) It was difficult to assemble the separate A and B chains into a mature form.
(3) The human insulin gene contains introns that *E. coli* cannot splice out.
(4) The insulin produced in bacteria triggered severe immune responses.

Q174 Match **List-I** with **List-II**.

List-I		List-II	
(A)	Lysozyme	(I)	Fungus
(B)	Cellulase	(II)	Plant cells
(C)	Chitinase	(III)	Proteins
(D)	Protease	(IV)	Bacteria

Choose the **correct** answer from the options given below:
(1) A-IV, B-II, C-I, D-III
(2) A-IV, B-I, C-II, D-III
(3) A-II, B-IV, C-I, D-III
(4) A-I, B-II, C-III, D-IV

Q175 Which of the following statements regarding a chemical synapse is/are **incorrect**?
(A) The arrival of an action potential at the presynaptic terminal directly causes neurotransmitter vesicles to fuse with the membrane.
(B) Neurotransmitters diffuse across the synaptic cleft and bind to specific receptors on the postsynaptic membrane.
(C) The response of the postsynaptic neuron can be either excitatory or inhibitory, depending on the neurotransmitter and the receptor type.
(D) After binding, neurotransmitters are typically removed from the synaptic cleft by reuptake, enzymatic degradation, or diffusion.
(1) A only (2) D only
(3) B and C (4) A and D

Q176 Which of the following shows the **correct** chronological order (from oldest to most recent)

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of periods in the Mesozoic era?
(1) Cretaceous → Jurassic → Triassic
(2) Jurassic → Triassic → Cretaceous
(3) Triassic → Jurassic → Cretaceous
(4) Triassic → Cretaceous → Jurassic

Q177 Read the following statements:
(A) Alpha-interferons used in cancer therapy are biological response modifiers.
(B) Immunotherapy in cancer primarily activates innate immunity alone.
(C) Cancerous cells just continue to divide giving rise to masses of cells called tumors.
(D) Sarcomas are cancers of epithelial tissue.
Which of the above statements is/are **correct**?
(1) A and B only (2) A and C only
(3) B and D only (4) A, B and C

Q178 Which of the following statements is **incorrect** regarding periodic abstinence?
(1) Sexual intercourse should be avoided during days 10–17 to prevent pregnancy.
(2) The fertile window is calculated based on the lifespan of the ovum (1 day) and sperm (3–5 days).
(3) This method provides 100% protection against pregnancy if followed strictly.
(4) Irregular cycles make this method less reliable.

Q179 Choose the **odd** one out with respect to factors contributing to normal fertility in males:
(1) Normal sperm morphology
(2) High motility
(3) Complete functional acrosome
(4) Ability to survive in high estrogenic Environment

Q180 Which of the following statements is **not** true about mammary glands?
(1) Glandular tissue is divided into 15–20 lobes.
(2) Alveolar cells synthesize milk.
(3) Mammary ampulla stores milk for immediate release.
(4) Milk ejection is caused by prolactin.

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TEST-01

NEET AIR (2027) [Special Test Series]

Answer Key

DATE : 13/08/2026

Q1	(3)
Q2	(1)
Q3	(2)
Q4	(1)
Q5	(4)
Q6	(4)
Q7	(2)
Q8	(4)
Q9	(3)
Q10	(2)
Q11	(1)
Q12	(3)
Q13	(2)
Q14	(4)
Q15	(3)
Q16	(1)
Q17	(1)
Q18	(3)
Q19	(2)
Q20	(2)
Q21	(1)
Q22	(3)
Q23	(4)
Q24	(3)
Q25	(2)
Q26	(1)
Q27	(2)
Q28	(2)
Q29	(2)
Q30	(2)
Q31	(2)
Q32	(1)
Q33	(2)
Q34	(2)
Q35	(3)
Q36	(3)
Q37	(2)
Q38	(1)
Q39	(3)
Q40	(3)
Q41	(1)
Q42	(3)
Q43	(1)
Q44	(4)
Q45	(3)
Q46	(4)
Q47	(3)
Q48	(4)
Q49	(1)
Q50	(4)
Q51	(3)
Q52	(4)
Q53	(3)
Q54	(3)
Q55	(2)
Q56	(4)
Q57	(2)
Q58	(2)
Q59	(2)
Q60	(1)
Q61	(2)
Q62	(1)

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Q63 (1)
Q64 (3)
Q65 (1)
Q66 (2)
Q67 (4)
Q68 (1)
Q69 (1)
Q70 (2)
Q71 (3)
Q72 (4)
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Q116 (3)
Q117 (1)
Q118 (3)
Q119 (4)
Q120 (2)
Q121 (2)
Q122 (4)
Q123 (3)
Q124 (1)
Q125 (2)
Q126 (1)

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Q127 (3)
Q128 (2)
Q129 (3)
Q130 (3)
Q131 (2)
Q132 (1)
Q133 (2)
Q134 (1)
Q135 (1)
Q136 (3)
Q137 (1)
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Q165 (2)
Q166 (2)
Q167 (1)
Q168 (2)
Q169 (4)
Q170 (2)
Q171 (1)
Q172 (3)
Q173 (2)
Q174 (1)
Q175 (1)
Q176 (3)
Q177 (2)
Q178 (3)
Q179 (4)
Q180 (4)

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Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

Particle thrown Vertically $\Rightarrow v_1 = \sqrt{u^2 + 2gh}$

Particle thrown horizontally $\Rightarrow$ $v_2 = \sqrt{u^2 + 2gh}$

Q2 Text Solution:

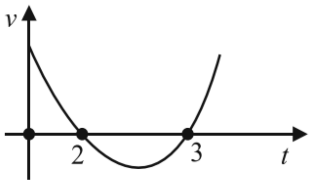
$\mu = \frac{v_d}{E} = \frac{2\times10^{-4}}{\frac{20}{10\times10^{-2}}}$ = 10^{-6} m^2 V^{-1} s^{-1}

Q3 Text Solution:

$v(t) = t^2 - 5t + 6$

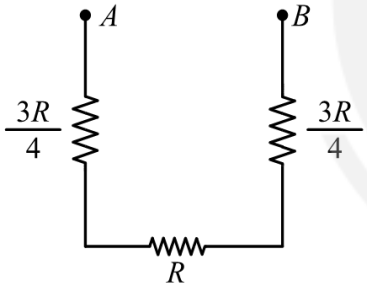
$v(t) = t^2 - 2t - 3t + 6$

$v(t) = (t-2)(t-3)$



At $t = \frac{5}{2}$s, it has nonzero speed and at t = 3s, it has zero speed i.e. it is slowing down.

Q4 Text Solution:



$R_{eq} = \frac{6R}{4} + R = \frac{5R}{2}$

Q5 Text Solution:

$W_1 = k[\ln(x)]_1^e = k(\ln(e) - \ln(1))$

$= k(1-0) = k$

$W_2 = k[\ln(x)]_e^{e^2} = k(\ln(e^2) - \ln(e))$

$= k(2-1) = k$

The ratio $W_1 : W_2$ is k : k or 1 : 1.

Q6 Text Solution:

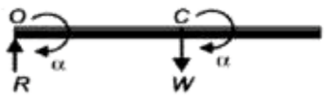
$U = \frac{1}{2}LI^2 = \frac{1}{2}L\frac{V^2}{R^2}$

$U' = \frac{1}{2}\frac{L}{16}\left(\frac{16V}{R}\right)^2$

$\frac{U}{U'} = \frac{1}{16} \Rightarrow U' = 16\,U$

Q7 Text Solution:

Let us suppose the reaction force experienced by the boy at the end be R and the instantaneous angular acceleration of the rod be α.



$\Sigma\tau_0 = I\alpha$

$W \cdot \frac{L}{2} = \frac{ML^2}{3} \cdot \alpha$

$\alpha = \frac{3g}{2L}$

$a_{com} = \frac{L}{2} \times \frac{3g}{2L} = \frac{3g}{4}$

$F_{net} = W - R = M \times a_{com}$

$W - R = M \times \left(\frac{3g}{4}\right)$

$R = \frac{W}{4}$

Q8 Text Solution:

AC Generator: It is a device which converts mechanical energy into electrical energy.

Galvanometer: It shows deflection when current passes through it, so it is used to show presence of current in the wire.

Transformer: It is used to step up or stepdown the voltage.

LED: It converts electrical energy to light energy.

Q9 Text Solution:

To get T we start from:

$T = 2\pi\sqrt{\frac{R^3}{GM}}$

Then we write mass in terms of density:

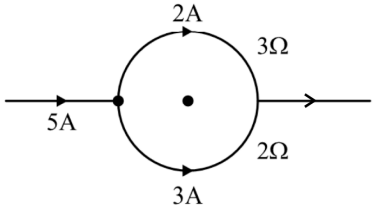
$M = \frac{4}{3}\pi R^3 d$

Substituting cancels R, leading finally to:

$T^2 = \frac{3\pi}{Gd}$

Q10 Text Solution:

Distribution of current



The magnetic field due to both will be in opposite directions

@shivamsh

$B_{net} = B_1 - B_2 = \frac{\mu_0(3)}{4(1)} - \frac{\mu_0(2)}{4(1)}$
$= \frac{\mu_0}{4} = \frac{4\pi\times10^{-7}}{4}$
$= \pi \times 10^{-7}\text{T}$

Q11 Text Solution:
$\lambda_m T = b.$
$T = \frac{2.9\times10^{-3}}{500\times10^{-9}} = \frac{2.9}{5} \times 10^4$
$= 0.58 \times 10^4 = 5800\text{K}$

Q12 Text Solution:
$F = \frac{kQ^2}{r^2}$
$F_1 = \frac{K\left(\frac{4Q}{5}\right)^2}{r^2} = \frac{16F}{25}$

Q13 Text Solution:
$a_1 = \frac{2M_1g - M_1g}{2M_1 + M_1} = \frac{g}{3}$
$a_2 = \frac{3M_1g - M_1g}{3M_1 + M_1} = \frac{g}{2}$
$\therefore \frac{a_1}{a_2} = \frac{2}{3}$

Q14 Text Solution:
$\frac{1}{C_{eq}} = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} = \frac{11}{6}$
$\Rightarrow C_{eq} = \frac{6}{11}\mu\text{F}$
$Q = C_{eq}V = \frac{6}{11} \times 22 = 12\,\mu\text{C}$
Charge appearing on positive plate of 3 μF capacitor = +Q = + 12 μC

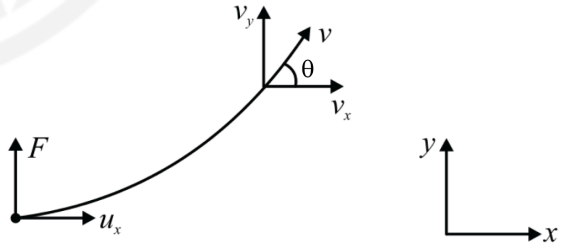
Q15 Text Solution:
During the oscillation of simple pendulum the restoring torque is provided by the component of weight of the bob.

Q16 Text Solution:
$m = +3 = \frac{-v}{u}$
$v = -3u$
$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$
$\Rightarrow \frac{-1}{3u} + \frac{1}{u} = \frac{1}{f}$
$\Rightarrow \frac{2}{3u} = \frac{1}{f}$
$\Rightarrow u = \frac{2f}{3} = \frac{2\times(-30)}{3} = -20\,\text{cm}$

Q17 Text Solution:
$f \propto \frac{1}{L}$. Ratio of frequencies 1 : 3 : 4.
Ratio of lengths $\frac{1}{1} : \frac{1}{3} : \frac{1}{4} = 12 : 4 : 3.$
Total parts $12 + 4 + 3 = 19$. Total length = 114 cm.
Length per part $= \frac{114}{19} = 6\,\text{cm}.$
$L_1 = 12 \times 6 = 72\text{cm}, \quad L_2 = 4 \times 6 = 24\text{cm}.$
$L_3 = 3 \times 6 = 18\text{cm}.$
Positions from one end:
First bridge at $L_1 = 72\text{cm}.$
Second bridge at
$L_1 + L_2 = 72 + 24 = 96\,\text{cm}.$

Q18 Text Solution:
$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$
$-\frac{1}{u} = \frac{0.5}{40} \Rightarrow u = -80\,\text{cm}$

Q19 Text Solution:
Efficiency $\eta = 1 - \frac{T_2}{T_1}.$
Given : $\eta = 20\% = 0.2$
$0.2 = 1 - \frac{T_2}{T_1} \Rightarrow \frac{T_2}{T_1} = 0.8 \Rightarrow T_2 = 0.8T_1$
Second Case:
The sink temperature is reduced to $(T_2 - 40)$, and new efficiency is 40% = 0.4:
$0.4 = 1 - \frac{T_2 - 40}{T_1}$
$\frac{T_2 - 40}{T_1} = 0.6 \Rightarrow T_2 - 40 = 0.6T_1$
Solve for T_1:
Substitute $T_2 = 0.8T_1$ into the second equation:
$0.8T_1 - 40 = 0.6T_1$
$0.2T_1 = 40$
$T_1 = \frac{40}{0.2} = 200\text{K}$

Q20 Text Solution:


$\tan\theta = \frac{v_y}{v_x}$
$v_x = u_x = 5\,\text{m/s}$
$v_y = u_y + a_y t = 0 + \frac{2\times10^{-6}\times10}{1\times10^{-6}} \times \left(\frac{1}{5}\right)$
$= 4\,\text{m/s}$
$\therefore \theta = \tan^{-1}\left(\frac{4}{5}\right)$

Q21 Text Solution:
Let the actual velocity of the rain is $\vec{v}_r = v_x\hat{i} - v_y\hat{j}$
The man's velocity is $\vec{v}_m = u\hat{i}$

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The relative velocity is

$\vec{v}_{r/m} = \vec{v}_r - \vec{v}_m = \left(v_x - u\right)\hat{i} - v_y\hat{j}$

The angle with the vertical is given by

$\tan\theta = \left|\frac{\text{horizontal component}}{\text{vertical component}}\right| = \frac{|v_x - u|}{v_y}$

Case 1: $u_1 = 4\text{m/s}, \theta_1 = 30^\circ$

$\tan\left(30^\circ\right) = \frac{4-v_x}{v_y} \Rightarrow \frac{1}{\sqrt{3}} = \frac{4-v_x}{v_y}$

$\Rightarrow v_y = \sqrt{3}\left(4 - v_x\right) \quad \ldots(1)$

Case 2: $u_2 = 8\text{m/s}, \theta_2 = 60^\circ$

$\tan\left(60^\circ\right) = \frac{8-v_x}{v_y} \Rightarrow \sqrt{3} = \frac{8-v_x}{v_y}$

$\Rightarrow v_y = \frac{8-v_x}{\sqrt{3}} \quad \ldots\left(2\right)$

Equating (1) and (2):

$\sqrt{3}\left(4 - v_x\right) = \frac{8-v_x}{\sqrt{3}}$

$\left(4 - v_x\right) = \frac{8-v_x}{\left(\sqrt{3}\right)^2}$

$12 - 3v_x = 8 - v_x$

$\Rightarrow 4 = 2v_x \Rightarrow v_x = 2\text{m/s}$

Substitute $v_x = 2$ back into (1):

$v_y = \sqrt{3}\left(4 - 2\right) = 2\sqrt{3}\text{m/s}$

Actual speed of rain

$\left|\vec{v}_r\right| = \sqrt{v_x^2 + v_y^2} = \sqrt{2^2 + \left(2\sqrt{3}\right)^2}$

$= \sqrt{4 + 12} = \sqrt{16} = 4\text{m/s}$

Q22 Text Solution:

$U_i = \frac{9\times10^9\times2\times2\times10^{-12}}{3\times10^{-2}} = 1.2\,\text{J}$

$U_f = \frac{9\times10^9\times2\times2\times10^{-12}}{10\times10^{-2}} = 36\times10^{-2} = 0.36\,\text{J}$

$\therefore\ |\Delta U| = |0.36 - 1.2| = 0.84\,\text{J}$

Q23 Text Solution:

$W = Fs\cos\theta = m\left(g + \frac{g}{2}\right) \times \left[\frac{1}{2} \times \frac{g}{2} \times t^2\right]$

$= \frac{3}{8}mg^2t^2$

Q24 Text Solution:

$\delta = (\mu - 1)A$

So, $\delta_C > \delta_B > \delta_A$

Q25 Text Solution:

$x_{cm} = \frac{A_1x_1 - A_2x_2}{A_1 - A_2}$

$x_{cm} = \frac{\left(\pi a^2\right)\left(0\right) - \left(\frac{a^2}{4}\right)\left(\frac{a}{2}\right)}{\pi a^2 - \frac{a^2}{4}}$

$= -\frac{a}{2(4\pi-1)} = -\frac{a}{8\pi-2}$

Given $x_{cm} = -\frac{a}{x}$, we have:

$x = 8\pi - 2 \approx 8(3.14) - 2 \approx 25.12 - 2$

$= 23.12$

To the nearest integer, $x = 23$.

Q26 Text Solution:

$\lambda = \frac{h}{p} \Rightarrow \frac{\lambda_p}{\lambda_e} = \frac{p_e}{p_p} = \sqrt{\frac{2Em_e}{2Em_p}}$

$\frac{\lambda_p}{\lambda_e} = \sqrt{\frac{m_e}{m_p}} = \sqrt{\frac{1}{1849}} = \frac{1}{43}$

Q27 Text Solution:

Poisson's ratio:

$\sigma = \frac{\text{magnitude of lateral strain}}{\text{magnitude of longitudinal strain}}$

Lateral strain = $0.30 \times 0.004 = 0.0012$

$= 1.2 \times 10^{-3}$

Q28 Text Solution:

U.V. rays have maximum frequency out of given options. Hence, they are most effective for emission of electron.

Maximum kinetic energy of photoelectron is directly proportional to frequency of incident light.

Q29 Text Solution:

By Pascal's Law, Pressure

$P_1 = P_2 \Rightarrow \frac{F_1}{A_1} = \frac{F_2}{A_2}.$

$\frac{50}{5} = \frac{F_2}{100} \Rightarrow F_2 = 10 \times 100 = 1000\text{N}.$

Mass $m = \frac{F_2}{g} = \frac{1000}{10} = 100\text{kg}.$

Q30 Text Solution:

$\Delta Q = \frac{\Delta\phi}{R}$

$\phi\left(2\right) = 5(2)^2 - 4\left(2\right) + 1$

$\phi(2) = (20 - 8 + 1)$ Wb = 13 Wb

$\phi(0) = 1$ Wb

$\Delta Q = \frac{(13-1)}{10}$ = 1.2 C

Q31 Text Solution:

$P = \frac{1}{3}\left(\frac{M}{V}\right)v_{rms}^2$

The mass (M) of the gas is constant (same mass of gas is transferred). The root mean square speed (v_{rms}) is constant.

This shows that: $P \propto \frac{1}{V}$

Initial State: $P_1 = \frac{Mv_{rms}^2}{3V} = P$

Final State (where $V_{new} = 4V$):

$P_{new} = \frac{Mv_{rms}^2}{3(4V)}$

@shivamsoi

$P_{new} = \frac{1}{4}\left(\frac{Mv_{rms}^2}{3V}\right)$

$P_{new} = \frac{P}{4}$

Q32 Text Solution:

Paramagnetic substance get weakly attracted towards a magnet. Its magnetic permeability is greater than 1 but does not have very high value.

Q33 Text Solution:

Given

$A = 0.25\text{m}, y = 0.25 \times 0.8 = 0.20\text{m}, KE = 14.4\text{J}$

Using $KE = \frac{1}{2}k\left(A^2 - y^2\right)$

$14.4 = \frac{1}{2}k\left(0.25^2 - 0.20^2\right)$

$= \frac{1}{2}k\left(0.0625 - 0.04\right)$

$= \frac{1}{2}k\left(0.0225\right) = 0.01125k.$

$k = \frac{14.4}{0.01125} = 1280\text{N/m}$

Q34 Text Solution:

$Y = \overline{\overline{A \cdot B}} + \overline{B}$

$= A \cdot B + \overline{B}$

Q35 Text Solution:

Heat lost by metal = Heat gained by (water + calorimeter).

$m_m s_m \Delta T_m = (m_w + W) s_w \Delta T_w.$

$50 \times s_m \times (150 - 50) = (100 + 10) \times 1 \times (50 - 25)$

$50 \times s_m \times 100 = 110 \times 25.$

$5000 s_m = 2750.$

$\therefore s_m = \frac{2750}{5000} = 0.55\text{cal/g}^\circ\text{C}.$

Q36 Text Solution:

For shortest wavelength the energy gap should be maximum.

The energy gap is maximum between

$n = 2$ to $n = 1$.

Q37 Text Solution:

Average Speed

$= \frac{\text{Total Distance}}{\text{Time}} = \frac{4A}{T} = 4\text{Af}$

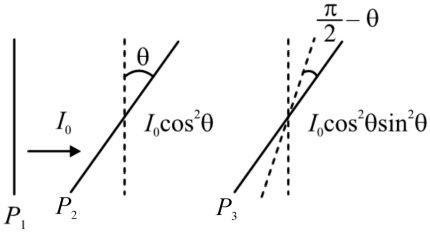
Given $A = 5.0\text{mm} = 0.005\text{m}$ and $f = 480\text{Hz}$:

$v_{\text{avg}} = 4 \times 0.005 \times 480$

$v_{\text{avg}} = 0.02 \times 480$

$\Rightarrow v_{\text{avg}} = 9.6\text{m/s}$

Q38 Text Solution:



Intensity of emergent light

$= I_0\left(\frac{\sin 2\theta}{2}\right)^2 = \frac{I_0}{4}(\sin 2\theta)^2$

Q39 Text Solution:

The system's total linear momentum is zero

$(\overrightarrow{P}_{\text{net}} = m\overrightarrow{v} - m\overrightarrow{v} = 0).$

Whenever the net linear momentum of a system is zero, its total angular momentum is independent of the choice of origin.

Since it's the equivalent of a "couple" in mechanics, the magnitude is simply the momentum of one particle multiplied by the perpendicular distance between their lines of motion.

$\left|\overrightarrow{L}\right| = \left(mv\right) \times d = mvd.$

Q40 Text Solution:

$\beta' = \frac{\lambda}{\mu} = \frac{0.6}{1.5} = 0.4\,\text{mm}$

Q41 Text Solution:

$\Delta M = \frac{E}{c^2} = \frac{18 \times 10^8}{(3 \times 10^8)^2}$

$= 2 \times 10^{-8}\,\text{kg}$

$= 2 \times 10^{-8} \times 10^3\,\text{g}$

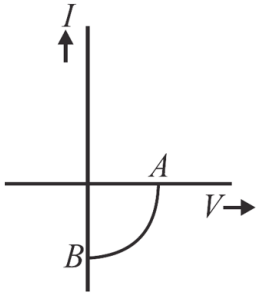
$= 2 \times 10^{-5}\,\text{g} = 20 \times 10^{-6}\,\text{g} = 20\,\mu\text{g}$

Q42 Text Solution:

$P = \frac{E_v^2 \cos\phi}{Z}$

$P = 3000 = \frac{(240)^2(0.75)}{Z} \Rightarrow Z = 14.4\ \Omega$

Q43 Text Solution:



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This given graph is *V-I* characteristic for solar cell where, point *A* represents open circuit voltage point *B* represents short circuit current.

Q44 Text Solution:

3.24 – 1.2 = 2.04

Final result should contain 1 decimal, i.e. 2.0

Q45 Text Solution:

Since $k = 4000\pi$, so wavelength

$= \frac{2\pi}{k} = \frac{2\pi}{4000\pi}$ m

= 0.5×10^{-3} m

Q46 Text Solution:

For nitrogen,

$$\text{Formal charge} = 5 - \frac{1}{2}(8) = +1$$

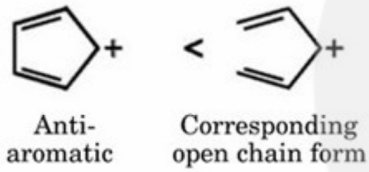
For oxygen,

$$\text{Formal charge} = 6 - 6 - \frac{1}{2}(2) = -1$$

Q47 Text Solution:

$$\text{Overall order of reaction} = 1.5 - 1 + 0 = 0.5$$

Q48 Text Solution:



Anti-aromatic

Corresponding open chain form

Q49 Text Solution:

For isotonic solutions $\pi_1 = \pi_2$

For 0.1 M glucose (non-electrolyte); $\pi = 0.1RT$

For 0.05 M NaCl (electrolyte produces Na^+ and Cl^- in solution); $\pi = 2 \times 0.05RT = 0.1RT$

\therefore Solution A and B are isotonic.

Q50 Text Solution:

Pressure does not have any significant effect on solubility of solids in liquids. It is because solids and liquids are highly incompressible.

Q51 Text Solution:

$$Ag_2O + H_2\overset{-1}{O}_2 \rightarrow 2Ag + H_2\overset{-2}{O} + \overset{0}{O}_2$$

H_2O_2 is oxidized to O_2 thus acts as a reducing agent.

Q52 Text Solution:

(1) and (2) are simple elimination products. (3) is ring expansion product.

Q53 Text Solution:

Total spin = no. of unpaired

$$e^- \times \left(\pm\frac{1}{2}\right) = 5 \times \left(\pm\frac{1}{2}\right) = \pm\frac{5}{2} \Rightarrow \pm 2.5$$

Q54 Text Solution:

Tetrahedral complexes having unidentate ligands do not show geometrical isomerism because the relative position of unidentate ligands attached to Central metal atom are same with respect to each other.

Q55 Text Solution:

According to molecular orbital theory, the energy level of the given molecules are

$C_2^{2-} \rightarrow \sigma 1s^2\ \sigma^*1s^2\ \sigma 2s^2\ \sigma^*2s^2\ \pi 2p_x^2\ \pi 2p_y^2\ \sigma 2p_z^2$

$\text{B.O.} = 1/2[10-4] = 3$

$He_2^+ \rightarrow \sigma 1s^2\ \sigma^*1s^1$

$\text{B.O.} = 1/2[2-1] = 1/2 = 0.5$

$NO \rightarrow \sigma 1s^2\ \sigma^*1s^2\ \sigma 2s^2\ \sigma^*2s^2\ \sigma 2p_z^2\ \pi 2p_x^2\ \pi 2p_y^2\ \pi^*2p_x^1$

$\text{B.O.} = 1/2[10-5] = 2.5$

$O_2^- \rightarrow \sigma 1s^2\ \sigma^*1s^2\ \sigma 2s^2\ \sigma^*2s^2\ \sigma 2p_z^2\ \pi 2p_x^2\ \pi 2p_y^2\ \pi^*2p_x^2\ \pi^*2p_y^1$

$\text{B.O.} = 1/2[10-7] = 1.5$

So, the correct order of their increasing bond order is

$He_2^+ < O_2^- < NO < C_2^{2-}$

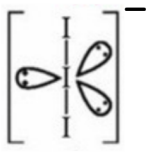
Q56 Text Solution:

(I) gives S_N1 product without rearrangement.

(II) gives S_N1 product with rearrangement.

(III) gives $E1$ product.

Q57 Text Solution:



Number of bond pairs around $I = 2$, Number of lone pairs around $I = 3$.

Q58 Text Solution:

As van't Hoff factor increases :

Relative lowering in vapour pressure increases.

i.e., vapour pressure decreases : $Y > X > Z$

Elevation in boiling point increases i.e., boiling point increases : $Y < X < Z$

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Depression in freezing point increases i.e., freezing point decreases : $Y > X > Z$
Osmotic pressure increases : $Y < X < Z$

Q59 Text Solution:

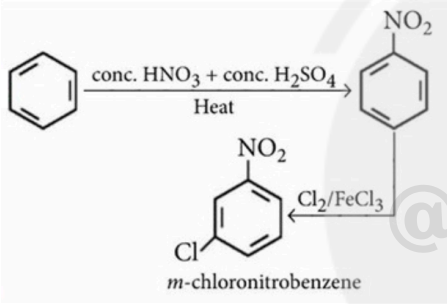
$$\underset{(X)}{Ph-NH_2} + \underset{(Y)}{CHCl_3} \xrightarrow{\text{alc. KOH}} Ph-N \rightrightarrows C$$

$$\underset{(Z)}{CH_3COCH_3} \xrightarrow{Cl_2/Ca(OH)_2} (CH_3COO)_2Ca + \underset{(X)}{CHCl_3}$$

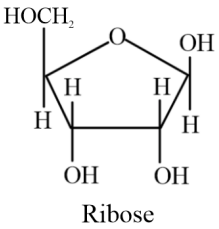
Q60 Text Solution:

The magnetic moment is determined by the number of unpaired electrons and is calculated by using the formula, $\mu = \sqrt{n(n+2)}$ where, n is number of unpaired electrons. It increases with increase in number of unpaired electrons. Hence Mn^{2+} will have higher magnetic moment than Cr^{2+}.

Q61 Text Solution:



Q62 Text Solution:

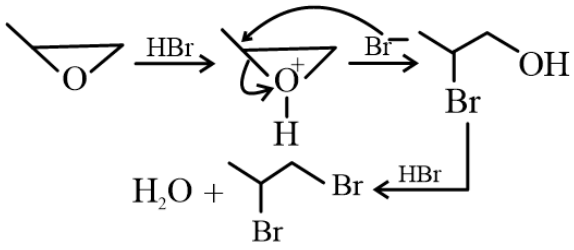


Q63 Text Solution:

This is attributed to the increasing stability of the lower species to which they are reduced.
MnO_4^- is reduced to Mn^{2+} which has stable half filled valence shell electronic configuration $[3d^5]$.
$Cr_2O_7^{2-}$ is reduced to Cr^{3+} which has half-filled t_{2g}^3 energy level of $3d$ orbitals in octahedral crystal field splitting. VO_2^+ is reduced to V^{3+} which has electronic configuration $[Ar]^{18}3d^24s^0$.

So the order of increasing stability of the reduced species is $Mn^{2+} > Cr^{3+} > V^{3+}$ and, therefore, the increasing order of oxidising power is $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$.

Q64 Text Solution:



Q65 Text Solution:

Carbon tetrachloride depletes the ozone layer. Chloroform is used in the production of freon refrigerant R-22.

Q66 Text Solution:

$[Pt(NH_3)_2Cl(NH_2CH_3)]Cl$
diamminechlorido(methanamine)platinum(II) chloride

Q67 Text Solution:

- Fullerenes have smooth structure without having dangling bonds.
- Hybridisation of carbon in diamond is sp^3.

Q68 Text Solution:

$$HA \rightleftharpoons H^+ + A^-$$
$$1 \quad 0 \quad 0$$
$$C(1-\alpha) \quad C\alpha \quad C\alpha$$

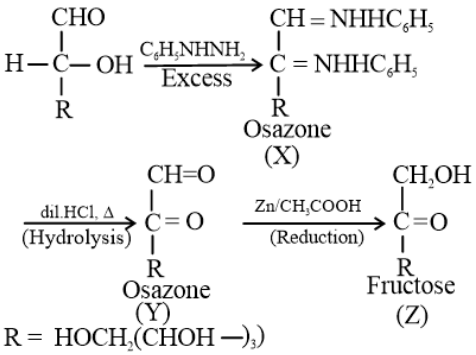
$K_a = C\alpha^2$
$10^{-4} = 10^{-2} \times \alpha^2 \Rightarrow \alpha = 0.1$
For HA, $i = 1 + \alpha = 1 + 0.1 = 1.1$

Q69 Text Solution:

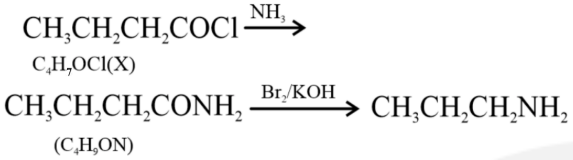
Neutral oxide	CO
Acidic oxide	SO_3
Basic oxide	MgO
Amphoteric oxide	As_2O_3

Q70 Text Solution:

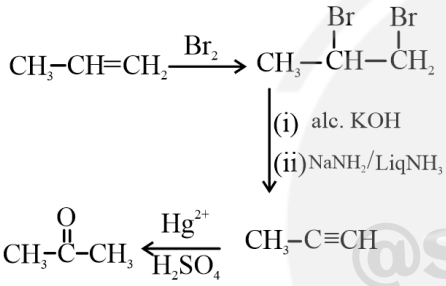
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Q71 Text Solution:



Q72 Text Solution:



Q73 Text Solution:

$$\underset{\text{Bond Energ (kj mol}^{-1})}{\text{H}-\text{H}} \quad \underset{433}{} + \underset{192}{\text{Br}-\text{Br}}$$

$$\rightarrow \underset{2\times364}{2\text{H}-\text{Br}}$$

Net energy released $= \sum H_R - \sum H_P$

$= (433 + 192) - 2 \times 364$

$= 625 - 728 = -103$ kJ

Q74 Text Solution:

Mn^{2+}	Group-IV
As^{3+}	Group-II B
Cu^{2+}	Group-II A
Al^{3+}	Group-III

Q75 Text Solution:

At $10^\circ C$ rise, rate increases by 2.

$\frac{r_{100^\circ C}}{r_{10^\circ C}} = 2^{\left(\frac{100-10}{10}\right)} = 2^9 = 512$ times

Q76 Text Solution:

Second order reaction	Unit of rate constant is $M^{-1}s^{-1}$
Zero order reaction	Thermal decomposition of HI on gold surface
First order reaction	Natural radioactive decay
Pseudo first order reaction	Inversion of cane sugar

Q77 Text Solution:

According to Gibb's equation,

$\Delta G = \Delta H - T\Delta S$ when

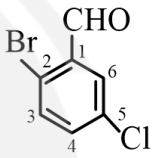
$\Delta G = 0, \Delta H = T\Delta S$

Given, $\Delta H = 40.63$ kJ mol^{-1} $= 40.63 \times 10^3$ J mol^{-1}

$\Delta S = 108.8$ J K^{-1} mol^{-1}

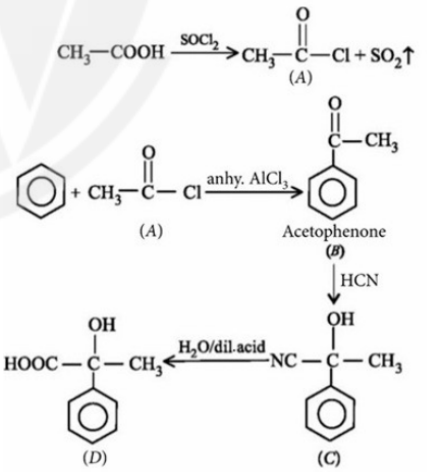
$\therefore T = \frac{\Delta H}{\Delta S} = \frac{40.63\times10^3}{108.8} = 373.43$ K

Q78 Text Solution:



2-bromo-5-chlorobenzaldehyde

Q79 Text Solution:



Q80 Text Solution:

$NaOH + CaO$	Decarboxylation of sodium salts of carboxylic acids
$V_2O_5 + Al_2O_3$	Dehydrogenation of alkanes
anhyd. $AlCl_3 + HCl$	Isomerisation of alkanes

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$H_2 + Pd/C$	Hydrogenation of alkynes

Q81 Text Solution:

Let the mass of the substance taken = w g

$\therefore \%C = \frac{12}{44} \times \frac{0.66}{w} \times 100 = \frac{18}{w}$

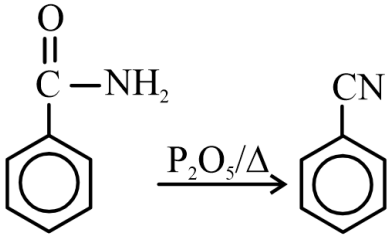
and $\%H = \frac{2}{18} \times \frac{0.36}{w} \times 100 = \frac{4}{w}$

Ratio of atoms,

$C : H = \frac{18}{w} \times \frac{1}{12} : \frac{4}{w} \times \frac{1}{1} = 3 : 8$

Hence, empirical formula = C_3H_8.

Q82 Text Solution:



Q83 Text Solution:

Mole fraction contain only mass term so it is independent of temperature. Sugar does not dissolve in benzene.

Q84 Text Solution:

Total no. of electrons in $n = 5$ are $2n^2$ i.e.,

$2 \times 5^2 = 50$

Thus, electrons having, $m_s = +\frac{1}{2} = \frac{50}{2} = 25$ which is equal to no. of orbitals.

Q85 Text Solution:

Adenine forms hydrogen bonds with thymine whereas cytosine forms hydrogen bonds with guanine. Due to specific pairing of bases the two strands are complementary.

Q86 Text Solution:

$Cu^{2+} + e^- \longrightarrow Cu^+; \Delta G_1 = -nFE_1{}^\circ$

$= -1 \times F \times E_1{}^\circ$

$Cu^+ + e^- \longrightarrow Cu; \Delta G_2 = -1 \times F \times 0.522$

$Cu^{2+} + 2e^- \longrightarrow Cu; \Delta G_3 = -2 \times F \times 0.34$

Now, $\Delta G_3 = \Delta G_1 + \Delta G_2$

$-2 \times F \times 0.34 = -FE_1{}^\circ - 0.522F$

$E_1{}^\circ = +0.158$ V

Q87 Text Solution:

Mass of organic compound = w g

Mass of magnesium pyrophosphate = w_1 g

$\underset{222g}{Mg_2P_2O_7} = \underset{62\ g}{2P}$

$\%$ of P $= \frac{62}{222} \times \frac{w_1}{w} \times 100$

$= \frac{62}{222} \times \frac{0.444}{0.248} \times 100 = \frac{2752.8}{55.056} = 50\%$

Q88 Text Solution:

Dipole moment μ is zero for symmetrical molecules i.e., $\mu_{CCl_4} = \mu_{CH_4} = 0$ but $\mu_{CHCl_3} > 0$.

Q89 Text Solution:

Electron withdrawing groups like $-NO_2$ facilitates nucleophilic substitution reaction in chlorobenzene.

Q90 Text Solution:

Riboflavin	Cheilosis
Thiamine	Beriberi
Pyridoxine	Convulsions
Ascorbic acid	Scurvy

Q91 Text Solution:

The chemical method developed by Har Gobind Khorana was instrumental in synthesising RNA molecules with defined combinations of bases (homopolymers and copolymers). It was George Gamow, a physicist, who argued that since there are only 4 bases and if they have to code for 20 amino acids, the code should constitute a combination of bases. He suggested that in order to code for all the 20 amino acids, the code should be made up of three nucleotides. Marshall Nirenberg's cell-free system for protein synthesis finally helped the code to be deciphered. Severo Ochoa enzyme (polynucleotide phosphorylase) was also helpful in polymerising RNA with defined sequences in a template independent manner (enzymatic synthesis of RNA). The elucidation of the *lac* operon was also a result of a close association between a geneticist, Francois Jacob and a biochemist, Jacque Monod. They were the first to elucidate a transcriptionally regulated system.

Q92 Text Solution:

The red thalli of most of the red algae are multicellular. Some of them have complex body

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organisation. The food is stored as floridean starch which is very similar to amylopectin and glycogen in structure. The members of rhodophyceae are commonly called red algae because of the predominance of the red pigment, r-phycoerythrin in their body. Majority of the red algae are marine with greater concentrations found in the warmer areas. They occur in both well-lighted regions close to the surface of water and also at great depths in oceans where relatively little light penetrates. They reproduce asexually by non-motile spores and sexually by non-motile gametes. The gametes are pyriform (pear-shaped) and bear two laterally attached flagella in brown algae. They vary in colour from olive green to various shades of brown depending upon the amount of the xanthophyll pigment, fucoxanthin present in them.

Q93 **Text Solution:**

In a standard Mendelian monohybrid cross, complete dominance results in a phenotypic ratio of 3:1 in the F_2 generation. If the ratio deviates from 3:1 (for example, becoming 1:2:1), it indicates Incomplete Dominance (where the heterozygote shows an intermediate phenotype, like pink flowers in snapdragons) or Co-dominance, rather than complete dominance.

Q94 **Text Solution:**

Most prokaryotic cells, particularly bacteria, have a complex cell envelope consisting of three layers: the outermost glycocalyx, followed by the cell wall, and then the plasma membrane.

Prokaryotic cells definitely possess genetic material. Prokaryotes do synthesize ATP. While they lack mitochondria, ATP synthesis occurs across the plasma membrane (or in specialized infoldings like mesosomes) during cellular respiration. Prokaryotes lack membrane-bound organelles (like the nucleus, ER, or Golgi). Ribosomes (70S) are the only "organelles" (non-membrane bound) found in their cytoplasm, shared with eukaryotes. Prokaryotic cells can be motile or non-motile. Motile bacteria possess thin filamentous extensions from their cell wall called flagella.

Q95 **Text Solution:**

Detritivores (e.g., earthworm) break down detritus into smaller particles. This process is called fragmentation. The detritivores are eukaryotic organisms. Decomposition is largely an oxygen-requiring process. Bacterial (prokaryotic organisms) and fungal (eukaryotic organisms) enzymes degrade detritus into simpler inorganic substances. This process is called as catabolism. The rate of decomposition is controlled by chemical composition of detritus and climatic factors. In a particular climatic condition, decomposition rate is slower if detritus is rich in lignin and chitin, and quicker, if detritus is rich in nitrogen and water-soluble substances like sugars. Temperature and soil moisture are the most important climatic factors that regulate decomposition through their effects on the activities of soil microbes. Warm and moist environment favour decomposition whereas low temperature and anaerobiosis inhibit decomposition resulting in build up of organic materials.

Q96 **Text Solution:**

In several species such as papaya, male and female flowers are present on different plants, that is each plant is either male or female (dioecy). This condition prevents both autogamy and geitonogamy. Continuous xenogamy maintains genetic diversity and prevents the accumulation of harmful homozygous recessive alleles, thus preventing inbreeding depression. In papaya, geitonogamy is physically impossible because a single plant cannot have both male and female reproductive parts.

Q97 **Text Solution:**

Ideally, when resources in the habitat are unlimited, each species has the ability to realise fully its innate potential to grow in number. Then the population grows in an exponential or geometric fashion.

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If in a population of size N, the birth rates (not total number but per capita births) are represented as b and death rates (again, per capita death rates) as d, then the increase or decrease in N during a unit time period t (dN/dt) will be

$dN/dt = (b - d) \times N$

Let $(b-d) = r$, then

$dN/dt = rN$

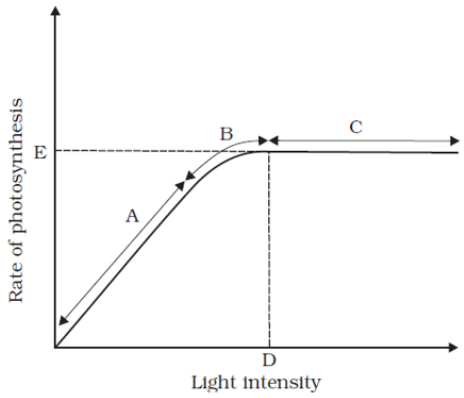
The above equation describes the exponential or geometric growth pattern of a population and results in a J-shaped curve when we plot N in relation to time.

Q98 Text Solution:

Slime moulds are saprophytic protists. The body moves along decaying twigs and leaves engulfing organic material. Red dinoflagellates cause red tides. In dinoflagellates, the cell wall has stiff cellulose plates on the outer surface. Instead of a cell wall, the euglenoids have a protein rich layer called pellicle which makes their body flexible. Diatoms are the chief 'producers' in the oceans.

Q99 Text Solution:

There is a linear relationship between incident light and CO_2 fixation rates at low light intensities (A and B). At higher light intensities (C), gradually the rate does not show further increase as other factors become limiting. What is interesting to note is that light saturation occurs at 10 per cent of the full sunlight (Point X). Hence, except for plants in shade or in dense forests, light is rarely a limiting factor in nature.



Graph of light intensity on the rate of photosynthesis

Q100 Text Solution:

As compared to the dicot root which have fewer xylem bundles, there are usually more than six (polyarch) xylem bundles in the monocot root. Pith is large and well developed. Monocotyledonous roots do not undergo any secondary growth.

Q101 Text Solution:

Since the number of chromosomes in the parent and progeny cells is the same, mitosis is also called as equational division. After a complete mitotic cell cycle (including Interphase where DNA is replicated and Mitosis where it is distributed), the daughter cells end up with the same amount of DNA as the parent cell had before it entered the S-phase (initial state).

Q102 Text Solution:

Biogas is a mixture of gases (containing predominantly methane) is produced by the microbial activity. Certain bacteria, which grow anaerobically on cellulosic material, produce large amount of methane along with CO_2 and H_2. These bacteria are collectively called methanogens, and one such common bacterium is *Methanobacterium*. These are a type of archebacteria.

Q103 Text Solution:

One turn of EMP pathway	2 ATP, 2 Pyruvate, 2 $NADH + H^+$
One turn of TCA cycle	1 ATP, 3 $NADH + H^+$, 1 $FADH_2$
Conversion of one pyruvate into acetyl CoA	1 $NADH + H^+$
Transport of 12 protons across F_0 particle of ATP synthase	3 ATP

Q104 Text Solution:

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Lactose is the substrate for the enzyme beta-galactosidase and it regulates switching on and off of the *lac* operon. Hence, it is termed as inducer. In the presence of an inducer, such as lactose or allolactose, the repressor is inactivated by interaction with the inducer. This allows RNA polymerase access to the promoter and transcription proceeds.

Q105 **Text Solution:**

Sickle-cell anaemia: The defect is caused by the substitution of Glutamic acid (GAG) by Valine (GUG) at the sixth position of the beta globin chain of the haemoglobin molecule.

Phenylketonuria : There are instances where a single gene can exhibit multiple phenotypic expression. Such a gene is called a pleiotropic gene. The underlying mechanism of pleiotropy in most cases is the effect of a gene on metabolic pathways which contribute towards different phenotypes. An example of this is the disease phenylketonuria, which occurs in humans. The disease is caused by mutation in the gene that codes for the enzyme phenyl alanine hydroxylase (single gene mutation). This manifests itself through phenotypic expression characterised by mental retardation and a reduction in hair and skin pigmentation.

Down's Syndrome : The cause of this genetic disorder is the presence of an additional copy of the chromosome number 21 (trisomy of 21), thus a total of 47 chromosomes.

Turner's Syndrome : Such a disorder is caused due to the absence of one of the X chromosomes, i.e., 45 with XO, thus a total of 45 chromosomes.

Q106 **Text Solution:**

The work of Mendel and others who followed him gave us an idea of inheritance patterns. However the nature of those `factors' which determine the phenotype was not very clear. As these `factors' represent the genetic basis of inheritance, understanding the structure of genetic material and the structural basis of genotype and phenotype conversion became the focus of attention in biology for the next century.

Q107 **Text Solution:**

There are certain limitations of ecological pyramids such as it does not take into account the same species belonging to two or more trophic levels. It assumes a simple food chain, something that almost never exists in nature; it does not accommodate a food web. Moreover, saprophytes are not given any place in cological pyramids even though they play a vital role in the ecosystem.

Q108 **Text Solution:**

Colombia located near the equator has nearly 1,400 species of birds while New York at 41° N has 105 species and Greenland at 71° N only 56 species. India, with much of its land area in the tropical latitudes, has more than 1,200 species of birds.

Q109 **Text Solution:**

Auxins help to initiate rooting in stem cuttings, an application widely used for plant propagation. 2, 4-D, widely used to kill dicotyledonous weeds, does not affect mature monocotyledonous plants. It is used to prepare weed-free lawns by gardeners.

Q110 **Text Solution:**

The meristem that occurs in the mature regions of roots and shoots of many plants, particularly those that produce woody axis and appear later than primary meristem is called the secondary or lateral meristem. They are cylindrical meristems. Fascicular vascular cambium, interfascicular cambium and cork-cambium are examples of lateral meristems. These are responsible for producing the secondary tissues. Both apical meristems and intercalary meristems are primary meristems.

Q111 **Text Solution:**

Dog and Gorilla both belong to the same Kingdom (Animalia), Phylum (Chordata), and Class (Mammalia). Dog belongs to order Carnivora and Gorilla belongs to order Primata.

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Therefore, they belong to different families as well.

Q112 Text Solution:

Axillary buds of stems may also get modified into woody, straight and pointed thorns. Thorns are found in many plants such as *Citrus*, *Bougainvillea*.

Q113 Text Solution:

Cilia	-	Meant for locomotion
Microvilli	-	Absorption of digested food materials
Microbodies	-	Present in both plant and animal cells
Lysosome	-	Possesses hydrolytic enzymes that digest biomolecules

Q114 Text Solution:

With the exception of yeasts which are unicellular, fungi are filamentous. The cell walls of fungi are composed of chitin and polysaccharides. Those that depend on living plants and animals are called parasites.

Q115 Text Solution:

RNA polymerase binds to promoter and initiates transcription (Initiation). It uses nucleoside triphosphates as substrate and polymerises in a template depended fashion following the rule of complementarity. It somehow also facilitates opening of the helix and continues elongation. Only a short stretch of RNA remains bound to the enzyme. Once the polymerases reaches the terminator region, the nascent RNA falls off, so also the RNA polymerase. This results in termination of transcription. The RNA polymerase is only capable of catalysing the process of elongation. It associates transiently with initiation-factor (σ) and termination-factor (ρ) to initiate and terminate the transcription, respectively.

Q116 Text Solution:

The technique of DNA Fingerprinting was initially developed by Alec Jeffreys. He used a satellite DNA as probe that shows very high degree of polymorphism. It was called as Variable Number of Tandem Repeats (VNTR). The technique, as used earlier, involved Southern blot hybridisation using radiolabelled VNTR as a probe. The VNTR belongs to a class of satellite DNA (not a polypeptide) referred to as mini-satellite.

A small DNA sequence is arranged tandemly in many copy numbers. The copy number varies from chromosome to chromosome in an individual. The numbers of repeat show very high degree of polymorphism. As a result the size of VNTR varies in size from 0.1 to 20 kb.

Q117 Text Solution:

Corn is pollinated by wind and *Vallisneria* is pollinated by water. Pollen grains coming in contact with the stigma is a chance factor in both wind and water pollination.

Q118 Text Solution:

The main plant body of the bryophyte is haploid. It produces gametes, hence is called a gametophyte. The sex organs in bryophytes are multicellular. The male sex organ is called antheridium. They produce biflagellate antherozoids. The female sex organ called archegonium is flask-shaped and produces a single egg. The antherozoids are released into water where they come in contact with archegonium. An antherozoid fuses with the egg to produce the zygote. Zygotes do not undergo reduction division immediately. They produce a multicellular body called a sporophyte.

Q119 Text Solution:

Chemiosmosis is primarily required for synthesis of ATP in chloroplasts. While NADPH + H^+ (not NADH + H^+) is produced during the light reactions of photosynthesis, its production is the result of the electron transport chain and the

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enzyme NADP reductase, not chemiosmosis itself. FNR stands for Ferredoxin-$NADP^+$ Reductase. It is an enzyme located on the stroma side of the thylakoid membrane. It facilitates the transfer of electrons from ferredoxin to $NADP^+$, reducing it to NADPH + H^+. Since it catalyzes a reduction reaction, it is indeed a type of reductase.

Q120 Text Solution:

- **Statement A is False:** Generally, **sepals** are green, leaf-like, and protect the flower in the bud stage, not petals.
- **Statement B is False: Petals** are brightly coloured to attract insects for pollination, not sepals.
- **Statement C is True:** When stamens in a flower remain free, the condition is termed **polyandrous**.
- **Statement D is True:** Each ovary bears one or more ovules attached to a flattened, cushion-like structure called the **placenta**.
- **Statement E is False:** There may be a variation in the length of filaments within a flower, as seen in ***Salvia*** and mustard. *Salvinia* is an aquatic fern (Pteridophyte), not an angiosperm.

Video Solution:

Q121 Text Solution:

In Anaphase I, the homologous chromosomes separate, while sister chromatids remain associated at their centromeres. Sister chromatids separate by splitting of centromere in Anaphase II. The nuclear membrane and nucleolus reappear in Telophase I and Telophase II.

Q122 Text Solution:

Streptokinase produced by the bacterium *Streptococcus* and modified by genetic engineering is used as a 'clot buster' for removing clots from the blood vessels of patients who have undergone myocardial infarction leading to heart attack.

Q123 Text Solution:

The physico-chemical approach to study and understand living organisms is called 'Reductionist Biology'. The concepts and techniques of physics and chemistry are applied to understand biology. The focus of this physico-chemical approach in biology is specifically to understand living organisms. While a dead cell still contains organic compounds, it no longer performs the dynamic metabolic processes that define life. The "role" of organic compounds is functional only within the context of a living, breathing, and reproducing system. Therefore, the study of dead cells is generally not the goal of this approach.

Q124 Text Solution:

Ferns	-	Pteridophytes
Mosses	-	Bryophytes
Naked seeds	-	Gymnosperms
Fruits	-	Angiosperms

Q125 Text Solution:

In conjoint type of vascular bundles, the xylem and phloem are jointly situated along the same radius of vascular bundles. In dicotyledonous stems, cambium is present between phloem and xylem. Such vascular bundles because of the presence of cambium possess the ability to form secondary xylem and phloem tissues, and hence are called open vascular bundles.

Q126 Text Solution:

Since respiration involves breakdown of substrates, the respiratory process has traditionally been considered a catabolic process and the respiratory pathway as a catabolic pathway.

Q127 Text Solution:

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Both DNA and RNA are able to mutate. In fact, RNA being unstable, mutate at a faster rate. There are two types of nitrogenous bases – Purines (Adenine and Guanine), and Pyrimidines (Cytosine, Uracil and Thymine).

Q128 Text Solution:

Maize (*Zea mays*) belongs to the family Poaceae. In most members of family Poaceae, sepals and petals are indistinct and form perianth, ovary is superior, fruit is single seeded, and scutellum is present in embryo.

Q129 Text Solution:

The infectious agent that causes potato spindle tuber disease is a viroid. Viroids (like viruses and prions) are sub-viral, infectious entities that do not have a cellular structure. They consist simply of a short strand of infectious RNA without a protein coat. Abnormally folded proteins are the infectious componenets of prions (which cause diseases like Mad Cow Disease), not viroids. Viroids are significantly smaller than bacteria and can easily pass through bacteria-proof filters. Viroids possess RNA. In fact, they are defined as low-molecular-weight, free RNA molecules. They only lack a protein coat (capsid).

Q130 Text Solution:

In the dihybrid cross, the phenotypes round, yellow; wrinkled, yellow; round, green and wrinkled, green appeared in the ratio 9:3:3:1. Based upon such observations on dihybrid crosses (crosses between plants differing in two traits) Mendel proposed a second set of generalisations that we call Mendel's Law of Independent Assortment.

Q131 Text Solution:

The kinetochore is a disc-shaped struture located on the centromere of a chromosome. Its primary function is to serve as an attachment site for spindle fibers (which are made of microtubules). While microtubules bind to the kinetochore during cell division, they are not a structural component of the kinetochore itself.

Q132 Text Solution:

Species composition and stratification are the two main structural features of an ecosystem.

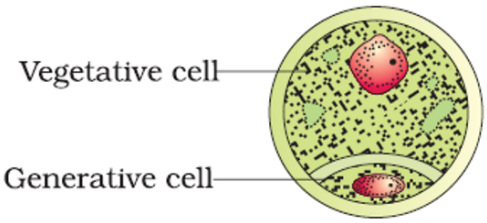
Productivity, decomposition, energy flow, and nutrient cycling are the four important components of an ecosystem. The storage and movement of nutrient elements through the various components of the ecosystem is called nutrient cycling; nutrients are repeatedly used through this process. Nutrient cycling is of two types—gaseous and sedimentary.

Q133 Text Solution:

Pollen grains are generally spherical measuring about 25-50 micrometers in diameter. It has a prominent two-layered wall. The hard outer layer (C) called the exine is made up of sporopollenin which is one of the most resistant organic material known. It can withstand high temperatures and strong acids and alkali. No enzyme that degrades sporopollenin is so far known. Pollen grain exine has prominent apertures called germ pores where sporopollenin is absent. Pollen grains are well-preserved as fossils because of the presence of sporopollenin.

The inner wall (D) of the pollen grain is called the intine. It is a thin and continuous layer made up of cellulose and pectin. The cytoplasm of pollen grain is surrounded by a plasma membrane. When the pollen grain is mature it contains two cells, the vegetative cell and generative cell. The vegetative cell is bigger, has abundant food reserve and a large irregularly shaped nucleus. The generative cell is small and floats in the cytoplasm of the vegetative cell. It is spindle shaped with dense cytoplasm and a nucleus. In over 60 per cent of angiosperms, pollen grains are shed at this 2-celled stage. In the remaining species, the generative cell divides mitotically to give rise to the two male gametes before pollen grains are shed (3-celled stage).

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A mature pollen grain

Q134 Text Solution:
Development in plants (i.e., both growth and differentiation) is under the control of intrinsic and extrinsic factors. The former includes both intracellular (genetic) or intercellular factors (chemicals such as plant growth regulators) while the latter includes light, temperature, water, oxygen, nutrition, etc.

Q135 Text Solution:
The genetic variation shown by the medicinal plant *Rauwolfia vomitoria* growing in different Himalayan ranges might be in terms of the potency and concentration of the active chemical (reserpine) that the plant produces.

Q136 Text Solution:
Nissl's granules are present in the cyton or cell body of the neuron, along with the usual cell organelles like nucleus, mitochondria, ER, and Golgi bodies. They are absent in the axon.

Q137 Text Solution:
In chordates, the CNS is dorsal, hollow, and single. In non-chordates, the nervous system is typically ventral, solid, and double. Chordates generally have a ventral heart, while non-chordates usually have a dorsal heart. Pharyngeal gill slits are present in chordates at least in some stage of life, not absent. Post-anal tail is a key chordate character.

Q138 Text Solution:
The cockroach thorax consists of three segments (prothorax, mesothorax, and metathorax). Each segment bears one pair of walking legs. Therefore, 3 pairs = 6 walking legs total.

Q139 Text Solution:
The water vascular system is indeed the most distinctive feature of echinoderms. It helps in locomotion, food capture, and respiration. The water vascular system does not function as an excretory system by filtering blood plasma. Echinoderms lack specialized excretory organs, and nitrogenous wastes mainly diffuse out through body surfaces, especially tube feet and papulae.

Q140 Text Solution:
All four:

- Heart sounds
- Blood pressure
- Pulse rate
- ECG

are of clinical diagnostic significance.
So, the number that are not of clinical diagnostic significance = 0.

Q141 Text Solution:
At high altitudes, oxygen availability is lower. In response, the body increases erythropoiesis (RBC production) to improve oxygen-carrying capacity.

Q142 Text Solution:
PCT, DCT, and collecting duct help regulate pH and ionic balance by selective secretion of H^+ ions and handling bicarbonate/ions.
Bowman's capsule mainly performs ultrafiltration and does not selectively secrete hydrogen ions for pH regulation.

Q143 Text Solution:

Liver	Secretes bile
Gall bladder	Temporarily stores bile
Pancreas	Produces digestive enzymes
Villi and Microvilli	Absorbs the final digested food

Q144 Text Solution:
The DNA fragments separate (resolve) according to their size through the sieving effect provided by the agarose gel, which acts as a molecular strainer.

Q145 Text Solution:

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Metameric segmentation (metamerism), where the body surface is distinctly marked out into segments or rings, is the hallmark of the Phylum Annelida (e.g., Earthworms).

Q146 Text Solution:

Plasma constitutes about 55% of blood (formed elements ≈ 45%)

Water content in plasma is ~90 – 92%

Remaining plasma components:

Proteins (6 – 8%)

Electrolytes, nutrients, hormones, etc.

Q147 Text Solution:

Transgenic mice are developed as in vivo models to test the safety of vaccines (like the polio vaccine) before they are administered to humans. If reliable, they can replace the use of monkeys.

alpha-1-antitrypsin is typically produced using transgenic sheep, not mice.

Human insulin is produced mainly by genetically engineered bacteria or yeast, not transgenic mice.

Human growth hormone is not extracted from transgenic mice for standard industrial use.

Q148 Text Solution:

Each human kidney contains about one million nephrons.

Blood leaves the glomerulus through the efferent arteriole.

Q149 Text Solution:

Ligases are the enzymes that exclusively catalyse the linking together of two molecules, playing a critical role in forming covalent bonds like C-O, C-S, C-N, and P-O.

Q150 Text Solution:

Recombinant therapeutics are designed to be human-compatible, so they usually do not trigger the kind of immune responses that substances obtained from non-human animals may cause. Recombinant human insulin made in bacteria is structurally identical to natural human insulin.

This structural identity is exactly why such recombinant products are less likely to cause immunological reactions.

Q151 Text Solution:

Somatostatin from the hypothalamus inhibits the release of growth hormone from the pituitary.

Q152 Text Solution:

Condoms	Prevent the physical meeting of ovum and sperms
Copper T (CuT)	Suppress sperm motility and lower fertilizing capacity
Oral pills	Inhibit ovulation and implantation while altering cervical mucus
Vasectomy	Block gamete transport entirely through surgical intervention

Q153 Text Solution:

Initially the population of Darwin's finches has medium-sized beaks because the island contains medium and large seeds.

After the disease:

- All plants with medium seeds disappear.
- Only large seeds remain.

Large seeds require stronger and larger beaks to crack them. Birds with slightly larger beaks will therefore:

- obtain food more efficiently,
- survive better,
- reproduce more.

Over generations, natural selection favors birds with larger beaks, causing the average beak size in the population to increase. This is an example of Directional Selection, where one extreme phenotype is favored.

Extinction is unlikely as long as there is a viable food source (large seeds) and enough genetic variation in the population to adapt.

If the environment changes significantly, the population must adapt or decline; no change is unlikely under high selection pressure.

Smaller beaks would be even less effective at cracking large, hard seeds therefore not favoured.

Q154 Text Solution:

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Frog's heart is three-chambered (two atria and one ventricle).

In male frogs, ureters do not act strictly as excretory ducts; they also carry sperms, so they are urinogenital ducts.

Frogs have a closed circulatory system and also possess a lymphatic system.

The partially digested food entering the duodenum from the stomach is called chyme.

Frogs lack external ears; they have tympanum visible externally.

Q155 Text Solution:

Rhinoviruses cause the common cold. Antibiotics (like streptomycin and amoxycillin mentioned in the other options) only work against bacteria.

There is currently no medicine that cures the common cold by killing the virus directly.

Since we cannot kill the virus with pills, the body must do the work itself. Rest and nutrition support the immune system, allowing it to clear the infection on its own.

Using antibiotics for a viral infection is not only ineffective but also contributes to antibiotic resistance, where bacteria evolve to survive the drugs meant to kill them.

Adding a pain killer helps with symptoms (like a headache or sore throat), but the presence of the antibiotic still makes this an incorrect treatment plan for a simple viral cold.

Q156 Text Solution:

Cell-mediated immunity (CMI) is carried out by T-lymphocytes and is responsible for graft rejection.

B-lymphocytes produce antibodies, but antibodies are proteins (immunoglobulins), not polysaccharides.

Q157 Text Solution:

Expiratory Capacity (EC) is the total volume of air a person can expire after a normal inspiration. It is mathematically the sum of Tidal Volume (TV) and Expiratory Reserve Volume (ERV).

Q158 Text Solution:

Plasmids and bacteriophages are widely used as cloning vectors in recombinant DNA technology.

They can replicate inside bacterial cells, which makes them suitable for carrying foreign DNA and producing multiple copies. This replication ability is exactly why they are used as cloning vectors.

Q159 Text Solution:

Sympathetic neural signals increase heart rate.

Epinephrine increases heart rate.

Nor-adrenaline also increases heart rate.

Parasympathetic neural signals decrease heart rate.

Q160 Text Solution:

Cephalochordates (like *Amphioxus*) use protonephridia with flame cells for excretion. Malpighian tubules are the characteristic excretory structures of insects (e.g., cockroaches).

Q161 Text Solution:

The hepatic portal vein carries nutrient-rich blood from the digestive tract (intestine) directly to the liver for processing before it is delivered into the general systemic circulation.

Q162 Text Solution:

Trophoblast	The outer layer of the blastocyst that attaches to the endometrium
Inner cell mass	An inner group of cells attached to the trophoblast that differentiates into the embryo proper
Chorionic villi	Finger-like projections appearing after implantation that interdigitate with uterine tissue
Morula	An early embryo consisting of 8 to 16 blastomeres

Q163 Text Solution:

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Sebaceous glands secrete sebum, which contains:

- Sterols
- Hydrocarbons
- Waxes

Lactic acid is a component of sweat, not sebum.

Q164 Text Solution:

For two pieces of DNA to ligate (stick together), their overhanging single-stranded ends—often called sticky ends—must be perfectly complementary (base-pair matching).

Let's look at the sticky ends each enzyme creates by cutting at the arrows:

1. RE1: Cuts between A and A. This leaves an overhang of AGCT.
2. RE2: Cuts between C and A. This also leaves an overhang of AGCT.
3. RE3: Cuts between G and A. This also leaves an overhang of AGCT.

Even though the recognition sequences are different for each enzyme, the specific sequence of the overhang is identical for all three:

- RE1 overhang: 5'-AGCT...
- RE2 overhang: 5'-AGCT...
- RE3 overhang: 5'-AGCT...

Because the overhangs are identical, they are compatible. RE1-digested DNA will find a perfect base-pairing match with DNA cut by RE2 or RE3. Since all three results in the same AGCT sticky end, they can all ligate to one another.

Q165 Text Solution:

TCT (thyrocalcitonin) and PTH (parathyroid hormone) together regulate blood calcium levels.

Addison's disease is caused by hyposecretion of adrenal cortex hormones, not hypersecretion.

Catecholamines stimulate glycogenolysis, causing hyperglycaemia.

Estrogen helps regulate female sexual behaviour.

Q166 Text Solution:

The primary advantage of genetic engineering over traditional hybridization is the ability to precisely isolate, clone, and transfer only the specific gene(s) of interest, avoiding the unwanted genetic baggage that accompanies whole-genome mixing in sexual reproduction.

Q167 Text Solution:

Ernst Haeckel gave embryological support for evolution based on similarities in vertebrate embryos.

Karl Ernst von Baer did not support the idea that embryos pass through the adult stages of other animals. In fact, von Baer opposed Haeckel's recapitulation idea.

Q168 Text Solution:

A muscle fibre is lined by the plasma membrane (sarcolemma) enclosing the sarcoplasm. It is considered a syncytium because the sarcoplasm contains many nuclei.

Q169 Text Solution:

In the mature Graafian follicle, the secondary oocyte gets surrounded by a non-cellular glycoprotein layer called the zona pellucida.

During fertilisation, the sperm interacts with the zona pellucida, which helps trigger changes that prevent polyspermy (entry of additional sperms).

Q170 Text Solution:

F-actin	Polymer of 'G' (globular) actins.
Tropomyosin	Runs close to the F-actins throughout their length.
Troponin	Masks the active binding sites for myosin.
Heavy Meromyosin (HMM)	Globular head with a short arm.

Q171 Text Solution:

The scapula is a large triangular flat bone situated in the dorsal part of the thorax between the second and the seventh ribs.

Q172 Text Solution:

Haemophilus influenzae (along with *Streptococcus pneumoniae*) is responsible for pneumonia, not the common cold. The common

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cold is caused by a group of viruses called rhinoviruses.

Q173 **Text Solution:**

The main challenge was assembling the insulin into its mature form. Human pro-insulin contains a C-peptide that bacteria cannot correctly excise and fold. Eli Lilly bypassed this by synthesizing the A and B chains separately in *E. coli* and chemically joining them via disulfide bonds later.

Q174 **Text Solution:**

Lysozyme	Bacteria
Cellulase	Plant cells
Chitinase	Fungus
Protease	Proteins

Q175 **Text Solution:**

Arrival of action potential at the presynaptic terminal does not directly cause vesicle fusion. It first opens voltage-gated Ca^{2+} channels, and Ca^{2+} influx triggers vesicle fusion.

Neurotransmitters diffuse across the synaptic cleft and bind to receptors on the postsynaptic membrane.

The postsynaptic response may be excitatory or inhibitory.

Neurotransmitters are removed by reuptake, enzymatic breakdown, or diffusion.

Q176 **Text Solution:**

The correct order of Mesozoic periods from oldest to most recent is: Triassic → Jurassic → Cretaceous.

Q177 **Text Solution:**

Alpha-interferons used in cancer therapy are biological response modifiers.

Immunotherapy does not activate only innate immunity; it can involve broader immune responses, especially adaptive immunity too.

Cancer cells keep dividing uncontrollably and form masses of cells called tumors.

Sarcomas arise from connective tissues, not epithelial tissue. Cancers of epithelial tissue are called carcinomas.

Q178 **Text Solution:**

In a typical 28-day cycle, avoiding intercourse during about days 10–17 helps prevent pregnancy.

The fertile window is estimated from the ovum's life (~24 hours) and sperm survival (3–5 days).

Periodic abstinence is not 100% effective, even if followed carefully.

Irregular menstrual cycles make this method less reliable.

Q179 **Text Solution:**

Normal sperm morphology, high motility, and a functional acrosome are all important for normal male fertility.

Survival in a high estrogenic environment is not a normal fertility requirement for sperm.

Q180 **Text Solution:**

Mammary glandular tissue is divided into 15–20 lobes. Alveolar cells synthesize and secrete milk.

The mammary ampulla acts as a temporary chamber before milk is released through the lactiferous duct. Prolactin stimulates milk production, while oxytocin causes milk ejection (let-down reflex).

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